THE CROWDED FRONTIER: STRATEGIC COMPETITION AND CONVERGENCE IN LLM LEADERBOARDS

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ABSTRACT

LLM evaluations are typically used to measure model capabilities, but once developers optimize against them, evaluations can also shape which capabilities developers choose to improve. We study the relationship between evaluation and model development in settings where developers' optimize towards win rate while minimizing a common function. We characterize when shared evaluation leads developers to converge toward similar capability profiles or towards specialized profiles. Similar production technologies are sufficient for similar capability profiles, while sufficiently strong comparative advantages induce specialization. Surprisingly, however, heterogeneous production technologies do not necessarily prevent monoculture, as even very different firms can respond similarly to a shared evaluation. This ambiguity ultimately limits what we can learn from leaderboards; static leaderboards tell us little about a firms underlying scaling law, while observing responses to changing evaluations provides additional identifying information. In response to this, we prove that variation in category weights can reveal more information than static leaderboards. Using longitudinal data from LLM Arena, we find evidence of both monoculture and specialization, with firms improving capabilities on which they lag behind leading models, consistent with pressure toward monoculture, while persistent firm-specific advantages across capabilities sustain specialization.. We show that these effects go beyond just firm-wide productivity, hinting at persistent differences between firms in relative strength across categories. In calibrated simulations, we show that varying the evaluation weights impact firms' investments and can give us better ability to recover information on the underlying technology, while having little impact on monoculture and specialization. Our results highlight that evaluation serves as a mechanism that both shapes innovation and determines what we can learn about the underlying technologies.

1 INTRODUCTION

Leaderboards are not only one of the main ways in which LLM capabilities are measured, but can also shape where developers focus their attention. For example, tournament-style leaderboards such as LLM Arena collect pairwise preferences from users for different pairs of LLMs. These pairs are then aggregated using Bradley-Terry preferences to construct a ranking of different models. Crucially, these evaluations do more than just measure what firms build; because firms strategically respond to them, evaluations can shape the direction of AI development. Therefore, an important question is what types of incentives evaluations might create, and the relationship between different types of evaluations and different types of incentives.

Because evaluation serves as a coordinating mechanism, a natural concern arises that a common leaderboard can lead to similar capability profiles or to monoculture. Monoculture is a prevalent idea within the economics literature, looking at how different market structures can lead to similarity in outcomes. In our setting, different evaluation mechanisms could lead to firms investing similarly, leading to a set of similar looking models. Our key question is when this occurs, and what role that evaluation plays.

While the relationship between monoculture and competition has been well-studied in the econ-cs literature, little work has examined how leaderboard-style competitions might impact these results.

We use techniques from the contest literature theory to study evaluation. From the contest literature theory, we use lottery Blotto games, along with models from information design and social choice theory (e.g., ranking) to better understand the equilibrium. Works in contest theory have studied similar styles of games, such as Blotto games and Lottery games. Work in the social choice looks at how the construction of leaderboards and voting can be improved, including the development of robust or pluralistic leaderboards. Within the field of evaluation, our work builds on work on strategic evaluation, including work on manipulation of LLM Arena, and on work looking at how leaderboards lead to unstable equilibria for model trainers. Unlike that model, we focus on how firms invest over time and with the key difference in our model being the presence of win-rate rather than rank-specific rewards.

In this work, we develop a model of competition among different AI firms to understand when evaluation can lead to monoculture . We model AI evaluation as a multidimensional contest in which firms allocate R&D investment across capabilities, their investments are transformed into model capabilities by potentially heterogeneous production functions F_i, and a common evaluator determines how those capabilities are rewarded. We ask three questions within this model. First, we ask what the incentives are for different firms, and when a common centralized evaluation leads homogenization as opposed to specialization. Second, we ask how firms respond to these incentives, and whether heterogeneous firms respond differently to the same capability gaps. Finally, we ask whether these observable pieces of information allow us to identify F_i and understand the underlying technologies behind these firms. Our results show that similar production functions lead to monoculture and comparative advantage leads to specialization, but that neither condition is necessary, with even very diverse or different production functions potentially leading to monoculture.

Our list of contributions is as follows:

1. We develop a model of tournament-style leaderboards, where different AI firms compete under a common cost function.
2. We show sufficient conditions for monoculture and specialization, and crucially show that neither condition is necessary, with monoculture arising even when production functions look wildly different.
3. We prove that production functions are not identifiable under a one-shot static picture of a leaderboard, while dynamic variations give us better identifiability
4. Using empirical data from Chatbot Arena, we show that firms have a mix of monoculture and specialization in practice.

2 Model and Preliminaries

We present our model of evaluation then discuss the definition of monoculture. We begin with N firms, each of which competes on K dimensions. Firm i invests amount $x_{i,k}$ in category k, resulting in an output $\boldsymbol{\theta}_i = F_i(\mathbf{x}_i) : \mathbb{R}_+^K \to \mathbb{R}_+^K$. Investment at level $\mathbf{x}$ results in a cost $c(\|\mathbf{x}_i\|_1)$, where c is a strictly increasing, differentiable, continuous, and unbounded cost function with $c(0) = 0$.

We focus on the setting where each firm is maximizing its "win-rate" or performance against other models. In practice, this corresponds to firms maximizing their performance on the leaderboard or maximizing their ELO, and we formally discuss this in Appendix A. Suppose that the K categories occur with frequencies $\mathbf{q} \in \Delta^{K-1}$ in the underlying distribution of user queries. The evaluator may reweight these categories through up- or downsampling, inducing an evaluation distribution $\mathbf{w} \in \Delta^{K-1}$. In the absence of such intervention, $\mathbf{w} = \mathbf{q}$; more generally, $\mathbf{w}$ captures the relative weight placed on each category by the evaluation procedure. Then firm i invests $\mathbf{x}_i$ to maximize the win-rate against other models:

$$\max_{\mathbf{x}_i} \frac{1}{N-1} \sum_{i' \neq i} \sum_k w_k \sigma(F_i(\mathbf{x}_i)_y - F_{i'}(\mathbf{x}_{i'})_y) - c(\sum_k x_{i,k}).$$

Here, σ denotes the win-probability according to some pairwise preference model (e.g., Bradley-Terry). We do not assume a specific preference model; we require only that σ is strictly increasing, twice differentiable, bounded in $[0, 1]$, and symmetric in the sense that $\sigma(x) + \sigma(-x) = 1$. The win probability captures the idea that human choices are inherently stochastic, depending on the relative

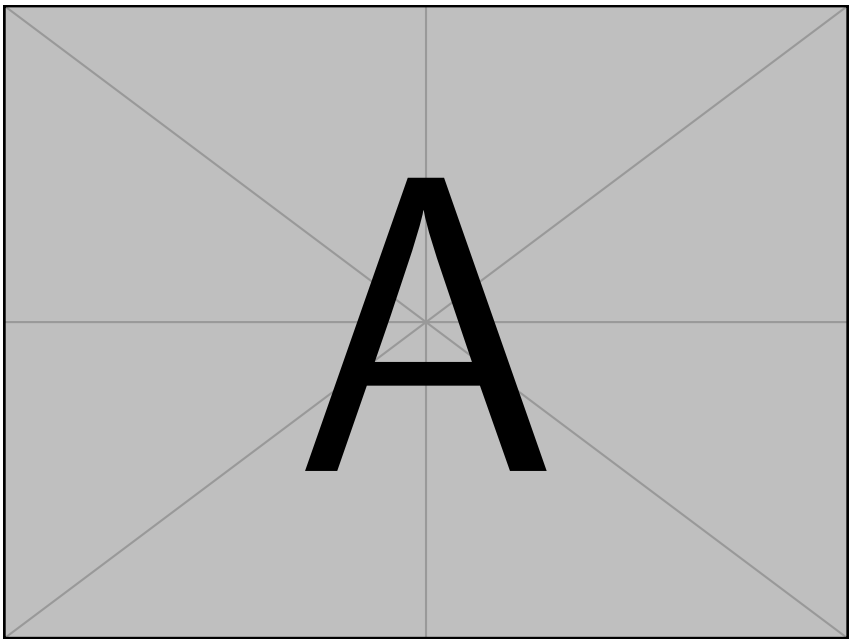


Figure 1:

difference between models. We also note that such a formulation is used by common ranking or leaderboard platforms as a way to capture model ranks. We additionally assume that firms play simultaneously; that is, they play $\mathbf{x}_i^* \in \arg\max_{\mathbf{x}_i \geq 0} u_i(\mathbf{x}_i, \mathbf{x}_{-i})$. We discuss whether such an equilibrium exists in Section 3. In Appendix E.1, we show that our model formulation is general; that it subsumes generalization in the cost function c to per-firm cost functions c_i. We prove this by first constructing a mapping g_i that maps between the cost functions ci and c, and let $\mathbf{z}_i = \frac{\mathbf{x}_i}{\|\mathbf{x}_i\|} g_i^{-1}(\|\mathbf{x}_i\|_1)$. Using this, we can define the alternate cost function $\tilde{F}_i(\mathbf{z}_i)$ as essentially scaling $\mathbf{z}_i$ based on $g^{-1}i$, which allows us to maintain $F_i(\mathbf{x}_i) = \tilde{F}_i(\mathbf{z}_i)$. We note that while the formulation is general, this reparameterization does not necessarily preserve properties such as concavity in the production function.

We next define monoculture in this setting as the tendency for firms to have similar outputs. We let $H_{i,i'} = \cos(\boldsymbol{\theta}_i, \boldsymbol{\theta}_{i'})$ be the pairwise monoculture for firms i and i', with $H_{i,i'} = 1$ indicating that $\boldsymbol{\theta}_i$ and $\boldsymbol{\theta}_{i'}$ are parallel. We can then define overall monoculture as $H = \frac{1}{\binom{N}{2}} \sum_{i,i'} H_{i,i'}$, with $H = 1$ indicating that all firms play parallel $\boldsymbol{\theta}_i$. Our measure captures similarity in firms' relative performance profiles across dimensions, abstracting away from differences in overall performance levels; this does not require that firms invest similarly however, due to differences in production functions. When $\boldsymbol{\theta}_i = \mathbf{0}$ and $\boldsymbol{\theta}_{i'} \neq \boldsymbol{\theta}_i$, we let $H_{i,i'} = 0$, while if $\boldsymbol{\theta}_i = \boldsymbol{\theta}_{i'} = \mathbf{0}$, then $H_{i,i'} = 1$. In Section 5 we discuss the importance of monoculture in terms of its impact on welfare, the key idea being that monoculture can directly impact welfare depending on user's cost for switching between models.

3 Existence of Pure Strategy Equilibria

We discuss when the model introduced in Section 2 exhibits pure-strategy equilibrium. We focus on pure-strategy Nash equilibria, which correspond to stable investment profiles in which no firm can benefit from unilaterally changing its investment. While mixed-strategy equilibria may exist even when no pure equilibrium exists, they require firms to randomize over investment decisions and do not correspond to a stable deterministic allocation of development effort.

3.1 Equilibrium with Two Firms

We first show when pure-strategy equilibria exists in a simplified setting with $K = 1$ and $N = 2$. Our optimization problem here is

$$u_1(x_1; x_2) = \sigma(f_1(x_1) - f_2(x_2)) - c(x_1), \qquad u_2(x_2; x_1) = \sigma(f_2(x_2) - f_1(x_1)) - c(x_2).$$

Suppose that, for every fixed action of the opposing firm, u_i is continuous and strictly concave in firm i's own action. Then each firm has a unique best response, which we denote by $\mathrm{BR}_1(x_2)$ and $\mathrm{BR}_2(x_1)$. A pair (x_1^*, x_2^*) is a Nash equilibrium if

$$x_1^* = \mathrm{BR}_1(x_2^*), \qquad x_2^* = \mathrm{BR}_2(x_1^*).$$

Equivalently, x_1^* is a fixed point of the composition of best responses,

$$\mathrm{BR}_1(\mathrm{BR}_2(x_1^*)) = x_1^*.$$

To give conditions for when this holds, we first note that u_i is continuous because σ, f_i, and c are all continuous. Since σ is bounded in $[0, 1]$ and c is increasing and unbounded, we can restrict each firm's action space to $[0, \bar{x}]$, where $\bar{x}$ satisfies $c(\bar{x}) = 1$. We additionally note that $h(x) = \mathrm{BR}_1(\mathrm{BR}_2(x)) - x$, and since $h(0) \geq 0$ and $h(\bar{x}) \leq 0$, and best response functions are continuous, then there exists a maximizer $x_1^* \in [0, \bar{x}]$. If u_i is additionally strictly concave in x_i, then each firm has a unique best response. Since u_i is continuous and the action space is compact, these unique best-response functions BR_1 and BR_2 are continuous.

We next give sufficient conditions for strict concavity. Let $z = f_1(x_1) - f_2(x_2)$,. Then

$$f_1''(x_1)\sigma'(z) + f_1'(x_1)^2\sigma''(z) - c''(x_1) < 0.$$

If c is convex and $f_1'(x_1) > 0$, a sufficient condition for concavity is that

$$f_1''(x_1)\sigma'(z) + f_1'(x_1)^2\sigma''(z) < 0,$$

Finally rearranging, and taking the supremum over z gives

$$\sup_z \frac{\sigma''(z)}{\sigma'(z)} < -\frac{f_1''(x_1)}{f_1'(x_1)^2} \forall x_1.$$

In Appendix B, we show that there exists scenarios where these conditions are not satisfied, and as a result, an equilibrium does not exist.

3.2 Existence of General Equilibria

We can now generalize when equilibria exist for general production functions F_i.

Theorem 1 (Equilibrium Existence)**.** *Suppose that each $g_{i,k}(\mathbf{x}_i) = [F_i(\mathbf{x}_i)]_k$ is twice continuously differentiable, c is convex, strictly increasing, continuous, and unbounded with $c(0) = 0$, and σ is twice continuously differentiable with $\sigma'(z) > 0$. Let*

$$\rho_\sigma := \sup_{z \in \mathbb{R}} \frac{\sigma''(z)}{\sigma'(z)}.$$

If

$$-\nabla^2 g_{i,k}(\mathbf{x}_i) \succeq \rho_\sigma \nabla g_{i,k}(\mathbf{x}_i) \nabla g_{i,k}(\mathbf{x}_i)^\top \qquad \forall\, \mathbf{x}_i,\ \forall\, i, k,$$

then the game admits a pure-strategy Nash equilibrium.

We can prove this through the Debreu-Fan-Glicksberg Theorem, which states that an equilibrium exists if u_i is quasi-concave and continuous with a compact convex strategy. In our setting, because $c(\mathbf{x}_i^*) \leq 1$, we can define the compact set $\mathcal{X}_i = \{\mathbf{x}_i \in \mathbb{R}_+^K : \|\mathbf{x}_i\|_1 \leq \bar{x}\}$. Computing the Hessian of u_i and invoking $-\nabla^2 g_{i,k}(\mathbf{x}_i) \succeq \rho_\sigma \nabla g_{i,k}(\mathbf{x}_i) \nabla g_{i,k}(\mathbf{x}_i)^\top$ completes the proof. We note that the requirements here are almost identical to that of the $N = 2$, $K = 1$ scenario, with just the terms themselves translated to higher dimensions.

4 When Does Monoculture Arise?

Given the existence of an equilibrium, we next ask when firms exhibit monoculture as opposed to differentiation. We give sufficient conditions for both. For monoculture, we show that similarity in production functions suffices, while for differentiation, we show that comparative advantage can induce firms to specialize. Notably, neither condition is necessary: monoculture can arise even when firms have substantially different production functions, as equilibrium behavior depends primarily on their local properties around the equilibrium.

4.1 Similar Production Functions Induce Monoculture

We begin by asking when monoculture arises at equilibrium. Intuitively, monoculture requires different firms to produce parallel outputs. A simple sufficient condition is therefore that firms with identical production functions choose identical investments. We show that such a symmetric equilibrium exists when production functions are identical:

Lemma 1 (Symmetric Firms)**.** *Suppose the conditions of Theorem 1 hold. If $F_i = F_{i'}$ for two firms i and i', then there exists a pure-strategy Nash equilibrium $\mathbf{x}^*$ such that $\mathbf{x}_i^* = \mathbf{x}_{i'}^*$, and so $\boldsymbol{\theta}_i^* = \boldsymbol{\theta}_{i'}^*$, implying that $H_{i,i'} = 1$.*

Moreover, if $F_i = F$ for every firm i, then there exists a symmetric pure-strategy Nash equilibrium satisfying $\mathbf{x}_1^* = \cdots = \mathbf{x}_N^*$, and at this equilibrium, $\theta_1^* = \cdots = \theta_N^*$, and so $H = 1$.

We prove this by showing that when $F_i = F_{i'}$, firms i and i' face identical optimization problems. When $F_i = F_{i'}$ and $\mathbf{x}_i = \mathbf{x}_{i'}$, firms i and i' face identical optimization problems and hence have the same best-response correspondence. Under the conditions of Theorem 1, this correspondence is nonempty, convex-valued, and upper hemicontinuous. Restricting attention to profiles in which firms i and i' select the same best response, we can apply Kakutani's fixed-point theorem to obtain an equilibrium satisfying $\mathbf{x}_i^* = \mathbf{x}_{i'}^*$.

Next, we note that strict equality is not necessary for monoculture. Rather, homogeneity degrades gracefully in the similarity between firms. In particular, if two firms are similar both in F_i and ∇F_i, then we guarantee near-monoculture.

To establish this, we first introduce the *pseudo-gradient* of the game, $G(\mathbf{x})$, which we define as

$$G(\mathbf{x}) = \left(-\nabla_{\mathbf{x}_1} u_1(\mathbf{x}), \ldots, -\nabla_{\mathbf{x}_N} u_N(\mathbf{x})\right),$$

where $\mathbf{x} = (\mathbf{x}_1, \ldots, \mathbf{x}_N)$ is the joint strategy profile. Here, the pseudogradient captures how each firm locally wants to modify its investment strategy, holding the strategies of other firms fixed. It provides a convenient way to characterize equilibrium behavior and how that behavior changes as production functions are perturbed. The key observation is that similarity in F_i and ∇F_i induces similarity in pseudogradients; strong monotonicity then translates this similarity in incentives into similarity in equilibrium investments.

Lemma 2 (Similarity in Production Functions)**.** *Suppose the conditions of Theorem 1 hold. Additionally, suppose that the production functions are L-Lipschitz, and that the pseudogradient G is μ-strongly monotone:*

$$\left(G(\mathbf{x}) - G(\mathbf{x}')\right)^\top (\mathbf{x} - \mathbf{x}') \geq \mu \|\mathbf{x} - \mathbf{x}'\|_2^2 \qquad \forall \mathbf{x}, \mathbf{x}'.$$

If firms i and i' have similar production functions in both value and gradient:

$$\sup_{\mathbf{x}_i \in \mathcal{X}_i} \|F_i(\mathbf{x}_i) - F_{i'}(\mathbf{x}_i)\|_2 \leq \epsilon, \qquad \sup_{\mathbf{x}_i \in \mathcal{X}_i} \|\nabla F_i(\mathbf{x}_i)_k - \nabla F_{i'}(\mathbf{x}_i)_k\|_2 \leq \epsilon \qquad \forall k.$$

then the equilibrium $\mathbf{x}^*$ satisfies

$$\|\mathbf{x}_i^* - \mathbf{x}_{i'}^*\|_2 \leq O\left(\frac{\epsilon}{\mu}\right), \qquad \|F_i(\mathbf{x}_i^*) - F_{i'}(\mathbf{x}_{i'}^*)\|_2 \leq O\left(\epsilon\left(1 + \frac{L}{\mu}\right)\right).$$

Moreover, if $\|\boldsymbol{\theta}_i^*\|_2, \|\boldsymbol{\theta}_{i'}^*\|_2 \geq m > 0$, then

$$H_{i,i'} \geq 1 - O\left(\frac{\epsilon^2}{m^2}\left(1 + \frac{L}{\mu}\right)^2\right).$$

We prove this by first constructing a counterfactual game where $F_i = F_{i'}$. We then show that $\|\widetilde{G}(\mathbf{x}) - G(\mathbf{x})\|_2 \leq O(\epsilon)$ because σ' is Lipschitz. Next, we use strong monotonicty to show that similarity in $G(\mathbf{x})$ implies similarity in $\mathbf{x}_i$. Finally, the Lipschitzness of F_i and bounds on cosine similarity lead to bounds on F_i and $H_{i,i'}$.

These results illustrate how similarity in production functions can generate monoculture because production functions are the primary source of heterogeneity across firms in our model. Intuitively, evaluation serves a coordinating role here; it presents the same incentives to different firms, making it possible for them to optimally respond in the same way. If two firms share a production function, then, when faced with the same investments from other firms, they face the same optimization problem. This critically relies on the conditions of Theorem 1, where utility is concave in each firm's own investment, so the set of optimal responses is convex and the two firms can respond identically to this common pressure If utility was non-concave, then firms facing the same incentives need not coordinate on the same response. In particular, multiple separated optima can arise, allowing otherwise identical firms to specialize in different directions rather than converging toward the same output profile.

4.2 Comparative Advantage Induces Differentiation

We next describe how comparative advantage can induce differentiation between firms. While Lemma 2 shows that similar production functions induce monoculture, the inverse is not true: substantially different production functions do not necessarily guarantee differentiation. Rather, differentiation can arise from comparative advantage, where firms have different relative strengths across dimensions. Without such differences, firms may still find it advantageous to pursue similar strategies. Without this, it's advantageous for firms to pursue similar strategies.

To illustrate what comparative advantage looks like in practice, we introduce a simple example and detail when differentiation arises.

Example 1. *Suppose $N = K = 2$ with production functions of the form $F_i([x_1, x_2]) = [a_i \log(x_1 + 1), b_i \log(x_2 + 1)]$. For simplicity, suppose the two dimensions receive equal evaluation weight, so that $w_1 = w_2 = 1/2$. Here firms differ based on their strengths for each dimension, controlled by a_i and b_i. We seek to characterize when firm $i = 1$ specializes and only invests in dimension $k = 1$, while firm $i = 2$ invests only in dimension $k = 2$. Suppose that $\mathbf{x}_1^* = [\alpha_1, 0]$ and $\mathbf{x}_2^* = [0, \alpha_2]$ for some $\alpha_1, \alpha_2 > 0$. Because $\alpha_1, \alpha_2 > 0$, the active coordinates satisfy the interior first-order conditions, while the zero coordinates satisfy the corresponding boundary inequalities. Under concavity of each firm's utility in its own investment vector, these KKT conditions are necessary and sufficient for optimality. For firm 1, these conditions are*

$$\frac{\partial u_1}{\partial x_{1,1}}(\mathbf{x}^*) = 0, \qquad \frac{\partial u_1}{\partial x_{1,2}}(\mathbf{x}^*) \leq 0,$$

while for firm 2 they are

$$\frac{\partial u_2}{\partial x_{2,2}}(\mathbf{x}^*) = 0, \qquad \frac{\partial u_2}{\partial x_{2,1}}(\mathbf{x}^*) \leq 0.$$

Expanding these conditions for firm 1 gives

$$\frac{\partial u_1}{\partial x_{1,1}}(\mathbf{x}^*) = \frac{1}{2}\sigma'(a_1 \log(1 + \alpha_1)) \, \frac{a_1}{1 + \alpha_1} - c'(\alpha_1),$$

and

$$\frac{\partial u_1}{\partial x_{1,2}}(\mathbf{x}^*) = \frac{1}{2}\sigma'(-b_2 \log(1 + \alpha_2)) \, b_1 - c'(\alpha_1).$$

Therefore, under concavity, $\mathbf{x}_1^* = [\alpha_1, 0]$ is optimal for firm 1 if and only if

$$\frac{1}{2}\sigma'(a_1 \log(1 + \alpha_1)) \, \frac{a_1}{1 + \alpha_1} = c'(\alpha_1),$$

and

$$\frac{1}{2}\sigma'(-b_2 \log(1 + \alpha_2)) \, b_1 \leq c'(\alpha_1).$$

Combining these two conditions yields the necessary inequality

$$\sigma'(a_1 \log(1+\alpha_1)) \frac{a_1}{1+\alpha_1} \geq \sigma'(-b_2 \log(1+\alpha_2))\, b_1.$$

Applying the analogous condition to firm 2 *and using the symmetry of* σ'*, we obtain*

$$\sigma'(b_2 \log(1+\alpha_2)) \frac{b_2}{1+\alpha_2} \geq \sigma'(a_1 \log(1+\alpha_1))\, a_2.$$

Multiplying the two specialization inequalities yields

$$\frac{a_1 b_2}{(1+\alpha_1)(1+\alpha_2)} \geq a_2 b_1,$$

or equivalently,

$$\frac{a_1 b_2}{a_2 b_1} \geq (1+\alpha_1)(1+\alpha_2).$$

Therefore, a necessary condition for specialization is sufficiently strong comparative advantage; firm 1 *must be relatively more productive in dimension* 1*, while firm* 2 *must be relatively more productive in dimension* 2*.*

More generally, specialization arises when different firms' production functions concentrate output on different dimensions. We formalize this by allowing production functions to have some spillover onto other dimensions, and show that differentiation degrades smoothly with the magnitude of these spillovers. We capture this through γ, which denotes the magnitude of these spillovers, with $\gamma = 0$ essentially capturing full specialization.

Lemma 3 (Specialized Production Functions Induce Differentiation)**.** *Consider two firms* i *and* i'*, and suppose there exist disjoint sets of dimensions* $S_i, S_{i'} \subseteq [K]$ *and some* $\gamma \in [0,1]$ *such that*

$$\left\| F_i(\mathbf{x}_i)_{S_i^c} \right\|_2 \leq \gamma \left\| F_i(\mathbf{x}_i)_{S_i} \right\|_2 \qquad \forall \mathbf{x}_i \in \mathcal{X}_i,$$

and

$$\left\| F_{i'}(\mathbf{x}_{i'})_{S_{i'}^c} \right\|_2 \leq \gamma \left\| F_{i'}(\mathbf{x}_{i'})_{S_{i'}} \right\|_2 \qquad \forall \mathbf{x}_{i'} \in \mathcal{X}_{i'}.$$

Then any equilibrium $\mathbf{x}^*$ *with* $\|\boldsymbol{\theta}_i^*\|_2, \|\boldsymbol{\theta}_{i'}^*\|_2 > 0$ *satisfies*

$$H_{i,i'} \leq \frac{2\gamma}{1+\gamma^2}.$$

In particular, if $\gamma = 0$*, then* $H_{i,i'} = 0$*.*

We prove this by showing that $\boldsymbol{\theta}_i^*$ and $\boldsymbol{\theta}_{i,S_i}^*$ are close in angle, and similarly for $\boldsymbol{\theta}_{i'}^*$ and $\boldsymbol{\theta}_{i',S_{i'}}^*$. As a result, $\boldsymbol{\theta}_i^*$ and $\boldsymbol{\theta}_{i'}^*$ are nearly orthogonal, which we can prove from trigonometric identities. This demonstrates that differentiation arises when firms' production functions are specialized along sufficiently different dimensions, with the degree of differentiation controlled by the extent of cross-dimensional spillovers. Notably, Lemma 2 and Lemma 3 provide only sufficient conditions for high and low monoculture, respectively. We next ask what can be said when neither condition is satisfied.

4.3 Necessary Conditions for Monoculture and Specialization

We next derive necessary conditions for monoculture and specialization. For monoculture, firms must not only be able to produce parallel output vectors, but must be able to locally support the same output direction under the competitive incentives induced by that direction. Because monoculture requires this compatibility only locally, this condition places relatively weak restrictions on the production functions.

To formalize this, fix a candidate monoculture direction $\mathbf{d} \in \mathbb{R}_+^K$. For any scales $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_N) \in \mathbb{R}_{++}^N$, define the corresponding marginal evaluation vector for firm i by

$$p_{i,k}(\mathbf{d}, \boldsymbol{\alpha}) = \frac{w_k}{N-1} \sum_{j \neq i} \sigma'((\alpha_i - \alpha_j) d_k)\,.$$

We say that $\mathbf{d}$ is locally supportable by firm i if there exist $\boldsymbol{\alpha} \in \mathbb{R}^N_{++}$ and $\mathbf{x}_i \in \mathbb{R}^K_+$ such that

$$F_i(\mathbf{x}_i) = \alpha_i \mathbf{d} \neq 0$$

and

$$\nabla F_i(\mathbf{x}_i)^\top \mathbf{p}_i(\mathbf{d}, \boldsymbol{\alpha}) \leq c'(\|\mathbf{x}_i\|_1)\mathbf{1},$$

with equality in every coordinate ℓ such that $x_{i,\ell} > 0$. That is, $\mathbf{d}$ is locally supportable if firm i can produce an output parallel to $\mathbf{d}$ at an investment vector satisfying the firm's first-order optimality conditions under the corresponding marginal evaluation incentives.

Proposition 1 (Local Compatibility is Necessary for Monoculture)**.** *For each firm i, define its set of locally supportable output directions as*

$$\mathcal{B}_i(w) = \left\{ \frac{\mathbf{d}}{\|\mathbf{d}\|_2} : \mathbf{d} \textit{ is locally supportable by firm } i \right\}.$$

If there exists a pure-strategy equilibrium with complete monoculture and nonzero outputs, then

$$\bigcap_{i=1}^{N} \mathcal{B}_i(w) \neq \emptyset.$$

The result follows directly from the first-order conditions. At a monoculture equilibrium, there exist $\mathbf{d}$ and $\boldsymbol{\alpha}$ such that $F_i(\mathbf{x}_i^*) = \alpha_i \mathbf{d}$ for every firm i. Optimality of $\mathbf{x}_i^*$ implies

$$\nabla F_i(\mathbf{x}_i^*)^\top \mathbf{p}_i(\mathbf{d}, \boldsymbol{\alpha}) \leq c'(\|\mathbf{x}_i^*\|_1)\mathbf{1},$$

with equality on every coordinate with positive investment. Hence the equilibrium direction is locally supportable by every firm, so $\mathbf{d}/\|\mathbf{d}\|_2 \in \bigcap_i \mathcal{B}_i(w)$ (full proof in Appendix E.6).

Specialization requires firms' output vectors to have disjoint support. Thus, a simple condition ruling out complete specialization is that each firm's production technology necessarily produces output across all capabilities. Under such broad cross-capability spillovers, no two nonzero output vectors can have disjoint support.

Proposition 2 (Positive Spillovers Rule Out Complete Specialization)**.** *Suppose that for every firm i and every investment vector $\mathbf{x}_i$ such that $F_i(\mathbf{x}_i) \neq 0$,*

$$[F_i(\mathbf{x}_i)]_k > 0 \qquad \forall k \in [K].$$

Then no pure-strategy equilibrium with nonzero outputs can exhibit complete specialization.

Proof. Suppose toward a contradiction that $\mathbf{x}^*$ is a pure-strategy equilibrium with nonzero outputs and complete specialization. For any two firms $i \neq j$, complete specialization requires

$$F_i(\mathbf{x}_i^*)^\top F_j(\mathbf{x}_j^*) = 0.$$

Since outputs are nonzero, the assumption implies

$$[F_i(\mathbf{x}_i^*)]_k > 0 \qquad \text{and} \qquad [F_j(\mathbf{x}_j^*)]_k > 0$$

for every capability $k \in [K]$. Therefore,

$$F_i(\mathbf{x}_i^*)^\top F_j(\mathbf{x}_j^*) = \sum_{k=1}^{K} [F_i(\mathbf{x}_i^*)]_k [F_j(\mathbf{x}_j^*)]_k > 0.$$

□

These results give simple necessary conditions at the two extremes: monoculture requires a common locally supportable direction, while complete specialization requires technologies capable of producing disjoint-support outputs. We next characterize equilibrium similarity more generally and derive bounds between these extremes.

4.4 Characterizing when Monoculture and Specialization Occur

While Lemma 2 demonstrates that similar production functions induce near-monoculture, we show that similarity in production functions is not necessary. In fact, we give an example in which production functions differ substantially, yet near-monoculture emerges at equilibrium.

Example 2. *Consider the scenario with $N = K = 2$ and $\sigma(z)$ as the sigmoid function. Let*

$$F_i(\mathbf{x}_i) = A_i \log(\mathbf{x}_i + \mathbf{1}),$$

where the logarithm is applied componentwise and $A_i \in \mathbb{R}_+^{2\times 2}$. Let

$$A_1 = \begin{pmatrix} 1.0 & 0.3 \\ 0.2 & 0.9 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0.4 & 0.8 \\ 0.7 & 0.1 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 0.35 & 0.65 \end{pmatrix}.$$

Next, let the cost $c(\|\mathbf{x}_i\|_1) = \frac{\gamma}{2}\|\mathbf{x}_i\|_1^2$, with $\gamma = 0.15$.

To characterize the equilibrium, we solve the corresponding KKT conditions. For every dimension receiving positive investment, the first-order necessary unctions unctionscondition is

$$\sum_{k=1}^{2} w_k \, \sigma'\big(\theta_{i,k} - \theta_{i',k}\big) \, \frac{A_{i,kj}}{1 + x_{i,j}} = \gamma(x_{i,1} + x_{i,2}),$$

while for dimensions receiving zero investment, the left-hand side is weakly less than the marginal cost. Numerically solving these conditions yields

$$x_1^* \approx (0.115,\ 0.601), \qquad x_2^* \approx (0.613,\ 0),$$

and the corresponding outputs are

$$\boldsymbol{\theta}_1^* \approx (0.250,\ 0.445), \qquad \boldsymbol{\theta}_2^* \approx (0.191,\ 0.335).$$

The resulting cosine similarity is $H_{1,2} \approx 0.99997$, and so the equilibrium is near-complete monoculture.

Monoculture only captures similarity in firms' outputs at equilibrium; away from equilibrium, their production functions may differ substantially. Thus, observing $H_{i,i'} \approx 1$ restricts only the outputs realized at equilibrium and leaves considerable freedom in the production functions off equilibrium. In particular, two production functions need only produce approximately parallel outputs at their equilibrium investments to generate monoculture.

With this in mind, we seek a more general characterization of when equilibrium behavior produces monoculture or specialization. Rather than comparing production functions directly, we characterize the set of output directions that each firm can optimally produce under the range of marginal evaluation incentives induced by competition. This allows us to bound equilibrium similarity for general N, K, and production functions.

Proposition 3 (Directional Characterization of Equilibrium Similarity)**.** *Suppose that the conditions of Theorem 1 are satisfied. Let $\sigma'(z) \in [\underline{s}, \bar{s}]$, on $z \in \mathcal{Z}$ containing all attainable output differences, with $\underline{s} > 0$.*

Define the set of marginal evaluation weights

$$\mathcal{P}(w) := \left\{ p \in \mathbb{R}_+^K : \underline{s} w_k \le p_k \le \bar{s} w_k \quad \forall k \right\}.$$

For each firm i, define its directional response set

$$\mathcal{D}_i(w) := \left\{ \frac{F_i(\mathbf{x}_i(p))}{\|F_i(\mathbf{x}_i(p))\|_2} : p \in \mathcal{P}(w),\ \mathbf{x}_i(p) \in \arg\max_{\mathbf{x}_i \in \mathcal{X}_i} \left\{ p^\top F_i(\mathbf{x}_i) - c(\|\mathbf{x}_i\|_1) \right\},\ F_i(\mathbf{x}_i(p)) \neq 0 \right\}.$$

Then, at every pure-strategy Nash equilibrium with nonzero outputs,

$$\inf_{\substack{d_i \in \mathcal{D}_i(w) \\ d_{i'} \in \mathcal{D}_{i'}(w)}} d_i^\top d_{i'} \le H_{i,i'} \le \sup_{\substack{d_i \in \mathcal{D}_i(w) \\ d_{i'} \in \mathcal{D}_{i'}(w)}} d_i^\top d_{i'}.$$

We prove this by showing that, for each firm i, the equilibrium first-order conditions coincide with those of

$$\max_{\mathbf{x}_i \in \mathcal{X}_i} \left\{ (\mathbf{p}_i^*)^\top F_i(\mathbf{x}_i) - c(\|\mathbf{x}_i\|_1) \right\}$$

for some $\mathbf{p}_i^* \in \mathcal{P}(w)$. By concavity, these conditions are sufficient, so $\mathbf{x}_i^*$ solves this optimization problem.

The sets $\mathcal{D}_i(w)$ provide a direct interpretation of the role of production functions. If $\mathcal{D}_i(w)$ and $\mathcal{D}_{i'}(w)$ contain only similar directions, then every equilibrium must exhibit high similarity. Conversely, if every direction in $\mathcal{D}_i(w)$ is nearly orthogonal to every direction in $\mathcal{D}_{i'}(w)$, then every equilibrium must exhibit specialization. Thus, what matters is not equality or dominance of production functions themselves, but whether they map the range of marginal evaluation incentives induced by competition into similar output directions.

Geometrically, complete specialization is more restrictive than complete monoculture. Under monoculture, all firms share a common normalized output direction in S_+^{K-1}, leaving $K-1$ continuous degrees of freedom. Under specialization, nonnegativity requires firms to have pairwise disjoint output supports. If firm i has support of size s_i, the resulting configuration has $\sum_i (s_i - 1) \leq K - N$ continuous degrees of freedom. Thus, complete monoculture has at least $N-1$ more degrees of freedom than complete specialization. Moreover, if $N > K$, complete specialization is impossible. We next translate this geometric difference into a comparison of the probability of near-monoculture and near-specialization.

Theorem 2 (Monoculture is More Generic than Specialization)**.** *Fix an evaluation rule (w, σ) and cost function c. Let*

$$D^* = (d_1^*, \ldots, d_N^*) \in \mathcal{Y} := (S_+^{K-1})^N$$

denote the normalized equilibrium output directions, where $d_i^* = \frac{\boldsymbol{\theta}_i^*}{\|\boldsymbol{\theta}_i^*\|_2}$. Suppose that D^* is distributed according to a continuous density on $\mathcal{Y}$ that is bounded above and below.

Let

$$\mathcal{M} := \{(d_1, \ldots, d_N) \in \mathcal{Y} : d_1 = \cdots = d_N\}, \quad \mathcal{S} := \left\{(d_1, \ldots, d_N) \in \mathcal{Y} : d_i^\top d_{i'} = 0 \quad \forall i \neq i'\right\},$$

denote the set of complete monoculture configurations and specialization configurations respectively.

For $\epsilon > 0$, let

$$\mathcal{M}_\epsilon := \{D \in \mathcal{Y} : \operatorname{dist}(D, \mathcal{M}) \leq \epsilon\}, \qquad \mathcal{S}_\epsilon := \{D \in \mathcal{Y} : \operatorname{dist}(D, \mathcal{S}) \leq \epsilon\}.$$

Then, when $N \leq K$, as $\epsilon \to 0$,

$$\frac{\Pr(D^* \in \mathcal{S}_\epsilon)}{\Pr(D^* \in \mathcal{M}_\epsilon)} = \Theta\left(\epsilon^{N-1}\right) \longrightarrow 0.$$

We can interpret D^* distribution as arising from variation in firms' production functions. For example, suppose that firms' production functions are drawn from some underlying family $\mathcal{F}$. Different draws of $(F_1, \ldots, F_N)$ induce different equilibrium output directions, generating a distribution D^* over $\mathcal{Y}$. Rather than impose assumptions on the distribution of production functions and the resulting equilibrium mapping, we directly assume that the induced distribution D^* has a continuous density that is bounded above and below. This captures settings in which variation in production functions generates full-dimensional variation in equilibrium output directions, rather than systematically concentrating outcomes on particular directions or lower-dimensional sets. In reality, this assumes that models non-zero output in all dimensions.

We prove this through a dimension-counting argument on the output space. Because the density of D^* is bounded above and below, probabilities of sufficiently small neighborhoods are proportional, up to constant factors, to their volumes. Complete specialization has $N-1$ additional codimensions relative to complete monoculture, and hence its ϵ-neighborhood shrinks by an additional factor of ϵ^{N-1}. Thus, absent forces that systematically concentrate equilibrium outputs on specialized boundary configurations, near-complete monoculture is generically more likely than near-complete specialization.

Taken together, these results show that monoculture is fundamentally a property of equilibrium output directions rather than of production functions themselves. Similar production functions are sufficient but not necessary for monoculture, while dominance alone does not determine whether firms converge or specialize. More generally, firms' directional response sets characterize which output directions can arise under competition. Finally, absent systematic concentration on specialized boundary configurations, the geometry of the output space makes near-complete monoculture more generic than near-complete specialization. We next ask what an evaluator can learn from observed firm outputs, and what aspects of these outcomes the evaluator can control.

5 Competitive Dynamics and Identification

Section 4 established that firms' production functions play a central role in determining whether equilibrium exhibits monoculture or specialization.. We next consider a dynamic setting in which firms make investments over multiple periods. Let

$$\mathbf{X}_{i,t} = \sum_{s=1}^{t} \mathbf{x}_{i,s}$$

denote firm i's cumulative investment through time t. At each timestep t, firms myopically choose $\mathbf{x}_{i,t}$ to maximize

$$\frac{1}{N-1} \sum_{i' \neq i} \sum_{k} w_k \sigma(F_i(\mathbf{X}_{i,t-1} + \mathbf{x}_{i,t})_k - F_{i'}(\mathbf{X}_{i',t-1} + \mathbf{x}_{i',t})_k) - c(\|\mathbf{x}_{i,t}\|_1).$$

Thus, investments accumulate over time, but firms optimize only their current-period utility.

5.1 Evolution of Firms Over Time

We first ask whether firms' outputs converge or diverge over time. In general, neither need occur: firms may remain separated because their cumulative investments diverge or because differences in their production technologies persist at scale. We show that when cumulative investment differences remain bounded and technological differences become negligible at scale, concavity causes relative performance differences to vanish.

Proposition 4 (Bounded Investment Gaps)**.** *Let $\mathbf{X}_i^t \in \mathbb{R}_+^K$ denote the cumulative investment of firm i at time t. Suppose that each coordinate of F_i is increasing and concave, with $F_i(\mathbf{0}) = \mathbf{0}$.*

Suppose cumulative investment gaps remain bounded:

$$\sup_t \|\mathbf{X}_i^t - \mathbf{X}_j^t\|_\infty < \infty \qquad \textit{for every } i, j,$$

while $\min_k X_{i,k}^t \to \infty$. Suppose further that production technologies are asymptotically equivalent:

$$\frac{\|F_i(\mathbf{x}) - F_j(\mathbf{x})\|_2}{\|F_j(\mathbf{x})\|_2} \to 0 \qquad \textit{as } \min_k x_k \to \infty$$

for every pair of firms i, j. Then firms bunch in relative performance:

$$\frac{\|F_i(\mathbf{X}_i^t) - F_j(\mathbf{X}_j^t)\|_2}{\max\{\|F_i(\mathbf{X}_i^t)\|_2, \|F_j(\mathbf{X}_j^t)\|_2\}} \to 0 \qquad \textit{for every } i, j.$$

The result separates two sources of persistent performance differences: investment and technology. Concavity implies that bounded differences in cumulative investment become negligible relative to output as scale grows, while asymptotic equivalence ensures that technological differences are also lower-order. Together, these forces imply frontier bunching (full proof in Appendix E.7).

Proposition 4 provides a dynamic analogue to the forces underlying monoculture and specialization. When technological differences become small relative to overall progress, firm performance bunches at the frontier. In contrast, persistent differences in production technologies can sustain relative advantages in particular capabilities. Thus, heterogeneous production functions can generate both broad catch-up and persistent capability-specific advantages over time.

5.2 Identifying Production Functions from Dynamic Data

Dynamic observations can also reveal information about firms' underlying production technologies. As Example 2 demonstrates, equilibrium similarity alone does not identify the underlying production functions. Even if an evaluator observes both a firm's investment and output, a single input-output observation provides only K restrictions on a production technology with potentially many more degrees of freedom. We formalize this for $F_i(\mathbf{x}_i) = A_i \log(\mathbf{x}_i + \mathbf{1} + \epsilon\mathbf{1})$, where a single observation provides only K restrictions on the K^2 entries of A_i.

Proposition 5 (Production Functions Are Not Identified from Equilibrium Outcomes)**.** *Suppose that firm i has production function*

$$F_i(\mathbf{x}_i) = A_i \log(\mathbf{x}_i + \mathbf{1} + \epsilon\mathbf{1}), \qquad A_i \in \mathbb{R}^{K\times K},$$

for some $\epsilon > 0$, where the logarithm is applied componentwise. Fix an observed equilibrium investment $\mathbf{x}_i^*$ and output $\boldsymbol{\theta}_i^*$. Then the set of production matrices consistent with this observation,

$$\mathcal{A}_i = \left\{ A_i \in \mathbb{R}^{K\times K} : A_i \log(\mathbf{x}_i^* + \mathbf{1} + \epsilon\mathbf{1}) = \boldsymbol{\theta}_i^* \right\},$$

is an affine space of dimension $K(K-1)$. Consequently, observing the equilibrium investments and outputs of all N firms leaves $NK^2 - NK = NK(K-1)$ degrees of freedom across their production matrices.

Proof. Let $z_i = \log(\mathbf{x}_i^* + \mathbf{1} + \epsilon\mathbf{1})$. Since $\epsilon > 0$, we have $z_i \neq 0$. The observed output imposes $A_i z_i = \boldsymbol{\theta}_i^*$. The map $A_i \mapsto A_i z_i$ is a linear map from $\mathbb{R}^{K\times K}$ to $\mathbb{R}^K$ with rank K. Hence the set of matrices satisfying this equation has dimension

$$K^2 - K = K(K-1).$$

Applying the same argument independently to each of the N firms gives $NK(K-1)$ remaining degrees of freedom. □

Repeated observations can provide the missing restrictions. In the dynamic setting, each distinct cumulative investment vector provides another measurement of the same production function. Variation in evaluation weights $\mathbf{w}$ is one way to induce additional variation in these investment states, but identification ultimately depends on the rank of the observed investment responses rather than on variation in $\mathbf{w}$ itself.

Proposition 6 (Identification Through Dynamic Variation)**.** *Suppose $F_i(\mathbf{x}_i) = A_i \log(\mathbf{x}_i + \mathbf{1} + \epsilon\mathbf{1})$ and the evaluator observes cumulative investments and outputs $\{(\mathbf{X}_i^t, \boldsymbol{\theta}_i^t)\}_{t=1}^T$. Let*

$$Z_i = \begin{bmatrix} \log(\mathbf{X}_i^1 + \mathbf{1} + \epsilon\mathbf{1}) & \cdots & \log(\mathbf{X}_i^T + \mathbf{1} + \epsilon\mathbf{1}) \end{bmatrix},$$

and let $r_i = \operatorname{rank}(Z_i)$. Then the set of production matrices A_i consistent with these observations has dimension $K(K - r_i)$. In particular, if $\operatorname{rank}(Z_i) = K$, then A_i is uniquely identified.

Proof. Each observation implies

$$A_i \log(\mathbf{x}_i^{*(t)} + \mathbf{1} + \epsilon\mathbf{1}) = \boldsymbol{\theta}_i^{*(t)}.$$

Stacking these equations gives $A_i Z_i = \Theta_i$. For each of the K rows of A_i, this system imposes r_i independent linear restrictions on K unknowns, leaving $K - r_i$ degrees of freedom. Thus the set of consistent matrices has dimension $K(K - r_i)$. If $r_i = K$, no degrees of freedom remain and A_i is uniquely identified. □

Thus, identification depends on variation in firms' realized investment states, not directly on variation in evaluation weights. Changing $\mathbf{w}$ can nevertheless facilitate identification by inducing firms to move in new investment directions. If changes in $\mathbf{w}$ generate only linearly dependent transformed investments, they provide no additional identifying information; if they generate K linearly independent responses, A_i is fully identified.

Together, these results show how repeated competition reveals the structure of firms' production technologies. When technological differences become lower-order and cumulative investment gaps

remain bounded, firm performance bunches at the frontier; persistent technological differences can instead sustain relative advantages. At the same time, observing firms repeatedly provides information unavailable from a single equilibrium: each independent investment-output observation further restricts the underlying production function. Dynamics therefore affect both how differences between firms evolve and how much those differences reveal about the technologies that generate them.

6 Leaderboard Competition in Practice

We use data from LLM Arena, OpenRouter, and the Open LLM Leaderboard to study how firms compete in real-world settings, whether their capabilities exhibit increasing monoculture, and what observed model trajectories can reveal about the underlying production functions.

6.1 Experimental Setup

To better understand how our theoretical predictions bore out in practice, we explore real-world datasets containing information on firm behavior and model competition. Our goal is to understand how firms' capability profiles evolve, whether their behavior exhibits patterns consistent with the mechanisms in our model, and what these observations can tell us about the underlying production functions. We rely on three sources of data, each of which gives us information on a different aspect of evaluation.

Leaderboards Our primary longitudinal data come from Chatbot Arena. We construct a panel of 215 per-category leaderboard snapshots from May 2023 through May 2026, covering 368 models from 60 organizations and 26 categories. The snapshots contain category-level Bradley–Terry ratings, rating variances, and vote counts. We use this panel to measure firms' capability profiles, changes across model releases, and whether subsequent improvements are associated with dimensions in which firms previously lagged their competitors. We also use Arena's style-controlled ratings and, where needed, raw battle data for supplementary analyses.

OpenRouter To measure realized model usage, we use daily OpenRouter traffic data from January 2025 through June 2026. The data cover 347 models from 39 organizations over 521 days. Each day, we have access to the token usage for the top-50 models used that day. We aggregate token shares to the model and firm levels to study whether increasing monoculture in evaluated capabilities is associated with changes in realized model usage and deployment concentration. OpenRouter primarily captures developer and long-tail API traffic and does not include all direct enterprise or first-party API usage, so we interpret it as a measure of an important segment of deployment rather than the full market.

Our datasets are observational and contain many potential confounds, so these experiments do not causally identify the mechanisms in our model. Instead, we use them to establish stylized facts about model competition and examine whether observed patterns are consistent with the mechanisms highlighted by our theory. Where possible, we test whether these patterns remain robust after accounting for observable confounds. Accordingly, our results should be interpreted as descriptive evidence about model competition rather than causal tests of firm behavior.

Our goal is to ask three key questions in this section:

1. **RQ1:** Do firms' capabilities converge toward monoculture over time, and what patterns accompany this convergence?
2. **RQ2:** Do models capabilities themselves converge over time; is the top getting more or less crowded?
3. **RQ3:** What can observed model trajectories reveal about the underlying production functions?

We primarily use the longitudinal Arena data for RQ1, combine repeated model releases with benchmark data for RQ2, and use counterfactual evaluation constructions for RQ3. OpenRouter provides a complementary view of whether patterns in evaluated capabilities are reflected in realized model usage.

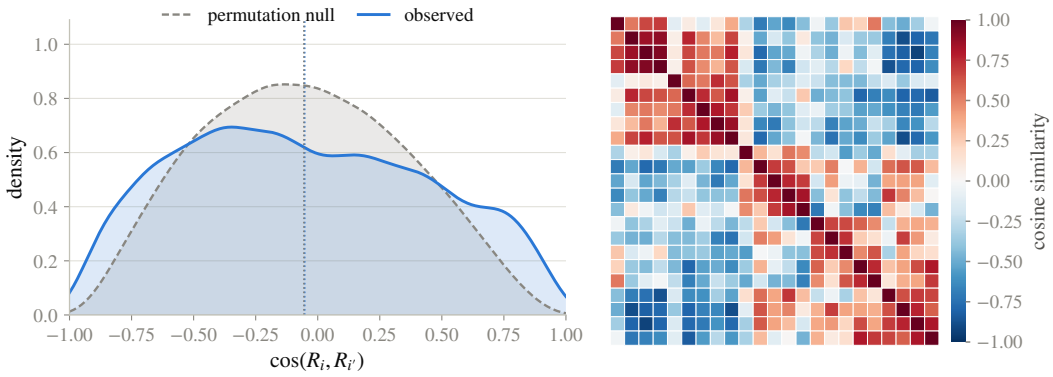


Figure 2:

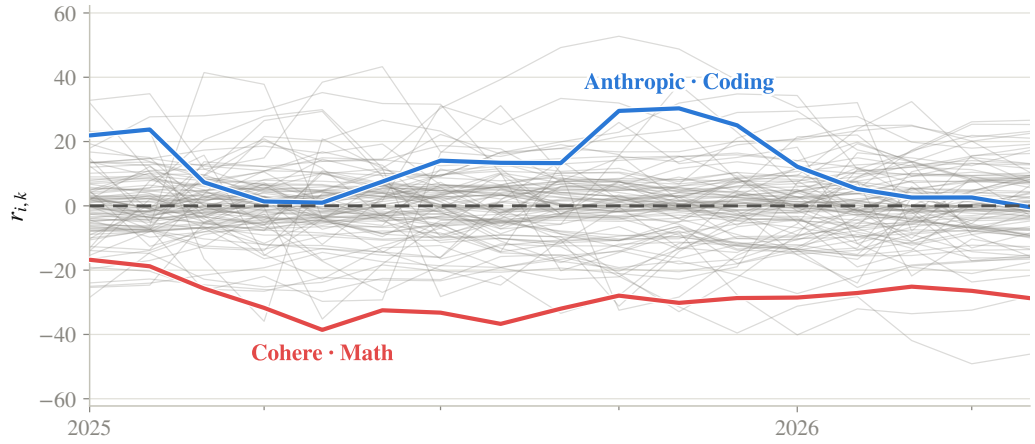


Figure 3:

6.2 RQ1: Are firms converging toward monoculture, and what drives this convergence?

Firm Similarities We first ask whether firms retain distinct capability profiles. Raw ELO vectors have pairwise cosine similarities near one, making them uninformative about relative strengths across categories. We therefore residualize each firm's category ratings by firm and category means, obtaining $r_{i,k} = \theta_{i,k} - \bar{\theta}_i - \bar{\theta}_k$, and examine both the cross-firm structure and temporal persistence of these residual profiles. [1]

Figure 2 shows that the aggregate distribution of pairwise residual similarity is close to a permutation null (mean cosine -0.053 vs. -0.056; $p = 0.106$), suggesting little systematic alignment across firms overall. However, the heatmap reveals substantial local structure: firms cluster into groups with similar relative strengths and weaknesses. For example, Alibaba, Baidu, DeepSeek, Google, OpenAI, and xAI form a cluster relatively stronger in Math and Creative Writing and weaker in Coding and English. Thus, firms do not share a single residual capability profile, but neither is differentiation idiosyncratic: several groups exhibit common patterns of comparative advantage.

We next ask whether these relative strengths persist over time. Figure 3 shows that many firm–category residuals fluctuate around zero, but some exhibit stable advantages or disadvantages. For

[1] Because residualization imposes cross-firm adding-up constraints, average pairwise residual cosine is mechanically negative and should not itself be interpreted as H. We instead use residual similarities to characterize clustering in relative capability profiles.

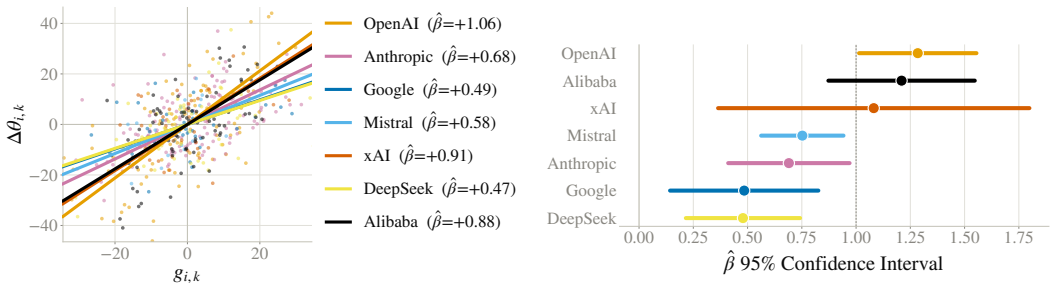


Figure 4:

Table 1:

	(A) Pooled	(B) Additive FE	(C) Firm × Cat.	(D) Model 3 + Cat. × Time FE	(E) + Cadence
$\hat{\beta}$ (Gap)	+0.354*** (0.060)	+0.314*** (0.075)	+0.334*** (0.079)	+0.338*** (0.081)	+0.354*** (0.094)
Firm FE	–	✓	–	–	–
Category FE	–	✓	–	–	–
Time (quarter) FE	–	✓	✓	–	–
Firm×Category FE	–	–	✓	✓	✓
Category×Time FE	–	–	–	✓	✓
Noise correction	–	–	–	–	–
N	520	520	520	520	456
Events	65	65	65	65	57

example, Anthropic persistently overperforms in Coding, while Cohere underperforms in Math; both patterns are statistically significant after correcting for temporal dependence and multiple testing. More broadly, 58 of 96 firm–category pairs exhibit significant nonzero mean residuals after FDR correction. Thus, firms retain meaningful and persistent comparative advantages even as their overall capability levels evolve.

Competitive Responses Persistent comparative advantage does not imply that firms specialize further. We next ask how firms respond when they fall behind the frontier in a particular category. For each flagship release, we relate the firm's subsequent improvement $\Delta\theta_{i,k}$ to its pre-release gap from the category leader $g_{i,k}$. Under specialization, firms may tolerate such gaps; under convergence, lagging categories should subsequently improve more.

Figure 4 shows a consistent gap-chasing pattern: firms improve more in categories where they previously lagged the frontier, with positive firm-level coefficients across all seven firms. A potential concern is mechanical regression to the mean, since noise in the outgoing model's rating enters both the measured gap and subsequent improvement. We therefore apply an errors-in-variables correction using the reported Bradley–Terry sampling variances. The corrected estimates remain positive for every firm and are similar or larger in magnitude. Thus, persistent comparative advantage coexists with a systematic force toward convergence: firms retain different strengths, but disproportionately improve the dimensions on which they lag.

Table 1 tests whether gap-chasing survives increasingly demanding controls. Our preferred specification absorbs persistent firm–category comparative advantage through firm×category fixed effects and common category-specific shocks through category×time fixed effects. The estimated gap coefficient remains remarkably stable, from 0.354 in the pooled specification to 0.338 in our preferred specification and 0.341 after additional cadence and sampling-noise corrections. Gap-chasing therefore appears distinct from both the persistent comparative advantages documented above and common improvements in particular capabilities.

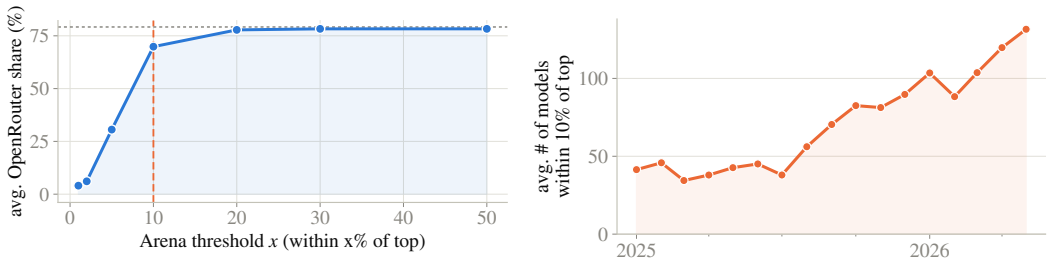


Figure 5:

Naveen: More robustness + experiments explaining this and removing possible confounds Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc fermentum vulputate rutrum. Duis id risus maximus, ultrices elit vel, sodales lectus. Vestibulum at egestas nibh. Nullam ac ultricies turpis, eu condimentum arcu. Etiam posuere libero eu velit varius convallis. Suspendisse quis lorem vel eros fringilla posuere sed at odio. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Sed mauris orci, blandit ac viverra sit amet, varius nec nisl. Etiam a odio cursus nunc eleifend vestibulum. Phasellus convallis mattis interdum. Nullam vehicula tincidunt enim sit amet condimentum.

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6.3 RQ2: Firm Crowding Over Time

We next investigate how changes in model performance correspond to real-world usage through an analysis of the OpenRouter dataset. We finally ask whether convergence near the evaluation frontier corresponds to greater substitutability in deployment. We match models between Arena and OpenRouter, covering 79% of monthly OpenRouter token volume on average. [2]

Figure 5 connects this convergence to deployment. In the left panel, OpenRouter usage is heavily concentrated among models near the Arena frontier: models within 10% of the best Arena performance account for approximately 70% of token volume, after which the curve largely plateaus. In the right panel, the number of models within this 10% frontier grows substantially over time, from 41.4 to 131.6 between January 2025 and June 2026.

Taken together, these results reveal a nuanced form of convergence. Firms retain persistent comparative advantages, but their subsequent releases disproportionately improve capabilities on which they lag the frontier. Over time, this process coincides with a growing set of near-frontier models, which account for the majority of observed OpenRouter usage. These patterns are consistent with our theoretical picture of evaluation-driven monoculture: competition need not eliminate all firm-level differentiation to create a common capability frontier toward which firms move.

Naveen: More thorough accounting for this phenomenon including another plot Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc fermentum vulputate rutrum. Duis id risus maximus, ultrices elit vel, sodales lectus. Vestibulum at egestas nibh. Nullam ac ultricies turpis, eu condimentum arcu. Etiam posuere libero eu velit varius convallis. Suspendisse quis lorem vel eros fringilla posuere sed at odio. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Sed mauris orci, blandit ac viverra sit amet, varius nec nisl. Etiam a odio cursus nunc eleifend vestibulum. Phasellus convallis mattis interdum. Nullam vehicula tincidunt enim sit amet condimentum.

[2] Unmatched volume is conservatively treated as outside the frontier threshold; see Appendix C for matching details.

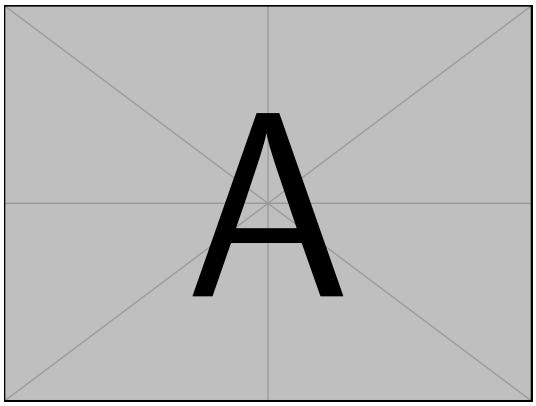


Figure 6:

F_i family	Diminishing returns	Firm×category heterogeneity	Persistent comp. advantage	Cross-category spillovers†
$a_i \log(1+x)\mathbf{1}$	✓	×	×	×
$A_i \log(1+\mathbf{x})$, diagonal A_i	✓	✓	✓	×
$A_i \log(1+\mathbf{x})$, general A_i	✓	✓	✓	✓
Flexible F_i	✓	✓	✓	✓

Table 2:

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6.4 RQ3: What can evaluation data reveal about firms' production functions?

We next ask what the observed capability trajectories reveal about firms' production functions.

We next ask which production-function families are capable of representing these empirical patterns. The data provide evidence for diminishing returns, firm and category heterogeneity, and persistent category-specific comparative advantage; our theory additionally allows cross-category spillovers, which these data do not directly identify. Table 2 summarizes which features can be represented by increasingly flexible production functions. A scalar firm-specific production function cannot capture the category-specific heterogeneity observed above. A diagonal A_i in $F_i(\mathbf{x}) = A_i \log(\mathbf{x} + \mathbf{1})$ captures heterogeneous returns across firm–category pairs, while a general A_i additionally permits cross-category spillovers. We therefore use this matrix-valued family as a flexible but tractable specification in the analysis that follows.

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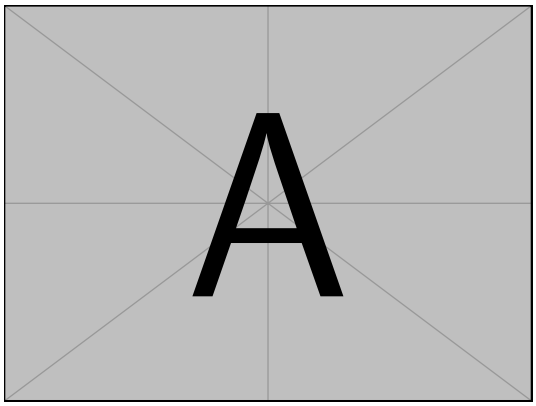


Figure 7:

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7 Implications for Evaluation

Conditions for Specialization and Monoculture Our theory highlights how properties of firms' production functions shape whether competition leads toward monoculture or specialization. In practice, however, we observe a mixture of the two: firms are neither perfectly aligned nor perfectly specialized. Importantly, monoculture and specialization need not be opposites. For example, subsets of firms may pursue similar capabilities while others maintain persistent advantages in particular categories.

One explanation for this intermediate structure is that AI development combines broadly accessible knowledge—through papers, open-source models, and shared engineering practices—with private, firm-specific knowledge and capabilities. Firm-specific advantages can generate persistent comparative advantage in some categories, while shared knowledge makes it possible for competitors to catch up and improve across a broad range of categories. The latter is particularly important: when advances are sufficiently transferable across firms, competitors can respond to one another across many categories rather than conceding categories in which they initially lag. Thus, the combination of heterogeneous firm-specific advantages and broadly accessible technological progress can generate the partial monoculture we observe in practice.

What role does evaluation play? Evaluation primarily plays a coordinating role. Public evaluations provide a common signal of which capabilities are rewarded and how firms compare, giving competitors a shared target toward which to optimize. Absent such a common signal, firms' incentives may be more dispersed, potentially encouraging differentiation or specialization (Raghavan, 2024). Evaluation instead makes competitors' strengths and weaknesses salient and creates incentives to improve along the same measured dimensions.

These incentives are reinforced by the production technology of general-purpose language models. Improvements in one capability often require investments that also improve other capabilities, making broad capability development relatively natural compared with complete specialization. Evaluation therefore need not create monoculture on its own. Rather, it can amplify the tendency toward broad, overlapping investment when firms face production technologies with substantial cross-category spillovers. In this sense, evaluation coordinates *where* firms compete, while the structure of production determines how strongly this competition translates into monoculture.

Limitations and Implications Our analysis combines a theoretical model with empirical patterns from LLM Arena and OpenRouter. The model necessarily abstracts from several features of real-world competition. We assume that firms optimize expected win rate rather than rank, an assumption that simplifies the strategic environment and affects equilibrium existence. We also focus on pure-strategy equilibria and assume that firms best respond to their competitors, whereas real firms may face uncertainty, adjustment costs, and imperfect information about competitors' strategies. Finally, our equilibrium results provide sufficient rather than necessary and sufficient conditions for existence.

Our empirical analysis is retrospective and observational. Although the patterns we document are consistent with the mechanisms highlighted by the model, establishing causal effects of evaluation is difficult because changes in evaluations, model capabilities, investment decisions, and the broader technological environment occur simultaneously. Our results should therefore be interpreted as identifying mechanisms through which evaluation can shape competitive investment, rather than as establishing that evaluation alone causes the observed degree of monoculture.

8 Related Work

Leaderboards Our work studies how preference-based leaderboards coordinate firms' incentives and, in turn, shape diversity. Leaderboards such as LLM Arena are online platforms that crowd-source language-model preferences, providing real-time and granular information about aggregate preferences, allowing for real-time and granular information on aggregate tastes (Chiang et al., 2024; Frick et al., 2025). Designing such leaderboards raises at least two challenges: aggregating heterogeneous preferences into a representative ranking (Rodriguez et al., 2021) and making that ranking robust to strategic manipulation. The first challenge is closely related to estimating top-k rankings from limited pairwise-comparison data (Chen & Suh, 2015; Rajkumar et al., 2015), while the aggregation of heterogeneous preferences can also be viewed through the lens of social choice (Lanctot et al., 2023). From a social-choice perspective, maximal lotteries provide one approach for aggregating heterogeneous preferences while satisfying desirable stability properties, although they may introduce additional computational complexity (Haghtalab et al., 2026; Lanctot et al., 2023).

For the latter goal of strategyproofness, leaderboards need to be resistant to adversarial manipulation by agents trying to influence rankings. Prior work has shown that naively constructed leaderboards built on top of Bradley-Terry preferences can be manipulated in a variety of ways, including through vote rigging (Min et al., 2025), strategic candidacy (Singh et al., 2025; Hays et al., 2026), and targeted manipulation (Huang et al., 2026). Whereas this literature primarily asks whether a leaderboard can be manipulated, we ask how the incentives created by evaluation shape firms' investment decisions and the resulting equilibrium structure. Building on Chen et al. (2026), we study firms that allocate costly investment across multiple categories to maximize their overall win rate.

Contest Theory Our model also connects to contest theory, with evaluation defining the contest through which firms compete. Contest theory studies how strategic agents expend costly effort to improve their probability of success (Borel, 1953; Friedman, 1958). Relevant models vary in the number of simultaneous battles (Konrad & Kovenock, 2009), the contest success function mapping effort into outcomes (Skaperdas, 1996), and whether outcomes are deterministic or stochastic (Osorio, 2013). Our model is closely related to multi-battle contest models (Tullock, 2001): firms allocate costly investment across categories, with pairwise success in each category determined by the function σ. Our focus differs from this literature in studying the similarity of equilibrium investment profiles across firms, and in particular, when contest incentives generate monoculture rather than specialization.

Strategic Machine Learning In many real-world settings, agents respond strategically to machine learning algorithms. In strategic classification, agents may modify their features in response to a classifier, requiring the learner to account for these endogenous responses (Hardt et al., 2016; Kleinberg & Raghavan, 2020). More broadly, performative prediction studies settings in which deploying a prediction rule changes the distribution on which future predictions are made (Perdomo et al., 2020). These strategic responses can also impose social costs, for example by inducing agents to expend additional effort simply to maintain favorable classifications (Milli et al., 2019). Our setting reverses the usual direction of strategic response: rather than individuals adapting to a de-

ployed model, model developers adapt their investments to an evaluation mechanism. We therefore study how evaluation itself induces strategic investment and how these responses shape equilibrium diversity (Blum & Hardt, 2015; Chen et al., 2026).

Competition and Monoculture Finally, our work connects to a growing literature on algorithmic monoculture and the effects of competition on diversity. Algorithmic monoculture describes settings in which multiple decision makers rely on similar algorithms, rankings, or predictions, causing their decisions to become correlated (Bommasani et al., 2022). Such correlation can reduce social welfare; even when adopting a common ranking is individually rational, widespread adoption can generate congestion or correlated errors that make the collective outcome worse (Kleinberg & Raghavan, 2021). This concern is particularly salient in domains such as hiring and college admissions, where a small number of algorithmic providers may influence decisions across many institutions (Bommasani et al., 2026; Peng & Garg, 2024).

In language models, even measuring monoculture is challenging because the appropriate counterfactual is unclear. For example, agreement across models about which examples are difficult may reflect monoculture, but may instead arise from shared properties of the underlying task (Kim et al., 2025; Jo et al., 2026). Prior work has shown that competition can act as an anti-coordination force, encouraging firms to differentiate (Raghavan, 2024). We identify a countervailing force; when competitors are evaluated against a common benchmark, evaluation can coordinate investment toward the same capabilities. Related work similarly shows that competition among increasingly capable models need not increase diversity and can, in some settings, reduce welfare (Jagadeesan et al., 2023). A related economics literature studies concentration among AI platforms. This work emphasizes forces such as complementary assets, compute requirements, and large fixed costs that may sustain a small number of firms (Korinek & Vipra, 2025; Azoulay et al., 2024), with potential consequences for diversity (Shaked & Sutton, 1982). Our notion of monoculture is distinct from market concentration: even a market with many firms may exhibit monoculture if those firms make similar investments. Closest to our empirical setting is Donahue & Raghavan (2026), which studies whether model aggregation through services such as OpenRouter produces homogenization. While both papers study diversity in model ecosystems, their mechanism is aggregation across existing models; ours is endogenous investment, asking whether common evaluation causes competing firms to develop similar capabilities.

9 Conclusion

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REFERENCES

Pierre Azoulay, Joshua L Krieger, and Abhishek Nagaraj. Old moats for new models: Openness, control, and competition in generative ai. Technical report, national bureau of economic research, 2024.

Avrim Blum and Moritz Hardt. The ladder: A reliable leaderboard for machine learning competitions. In *International Conference on Machine Learning*, pp. 1006–1014. PMLR, 2015.

Rishi Bommasani, Kathleen A Creel, Ananya Kumar, Dan Jurafsky, and Percy S Liang. Picking on the same person: Does algorithmic monoculture lead to outcome homogenization? *Advances in neural information processing systems*, 35:3663–3678, 2022.

Rishi Bommasani, Sarah H Bana, Kathleen A Creel, Dan Jurafsky, and Percy Liang. Algorithmic monocultures in hiring. In *The 2026 ACM Conference on Fairness, Accountability, and Transparency*, pp. 6351–6382, 2026.

Emile Borel. The theory of play and integral equations with skew symmetric kernels. *Econometrica: journal of the Econometric Society*, pp. 97–100, 1953.

Yatong Chen, Guanhua Zhang, and Moritz Hardt. Leaderboard incentives: Model rankings under strategic post-training. *arXiv preprint arXiv:2603.08371*, 2026.

Yuxin Chen and Changho Suh. Spectral mle: Top-k rank aggregation from pairwise comparisons. In *International conference on machine learning*, pp. 371–380. PMLR, 2015.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E Gonzalez, et al. Chatbot arena: An open platform for evaluating llms by human preference. *arXiv preprint arXiv:2403.04132*, 2024.

Kate Donahue and Manish Raghavan. Impacts of aggregation on model diversity and consumer utility. *arXiv preprint arXiv:2602.23293*, 2026.

Evan Frick, Connor Chen, Joseph Tennyson, Tianle Li, Wei-Lin Chiang, Anastasios N Angelopoulos, and Ion Stoica. Prompt-to-leaderboard. *arXiv preprint arXiv:2502.14855*, 2025.

Lawrence Friedman. Game-theory models in the allocation of advertising expenditures. *Operations research*, 6(5):699–709, 1958.

Nika Haghtalab, Ariel D Procaccia, Han Shao, Serena Lutong Wang, and Kunhe Yang. Pluralistic leaderboards. *arXiv preprint arXiv:2606.02547*, 2026.

Moritz Hardt, Nimrod Megiddo, Christos Papadimitriou, and Mary Wootters. Strategic classification. In *Proceedings of the 2016 ACM conference on innovations in theoretical computer science*, pp. 111–122, 2016.

Chris Hays, Rachel Li, Bailey Flanigan, and Manish Raghavan. Strategic candidacy in generative ai arenas. *arXiv preprint arXiv:2603.26891*, 2026.

Jenny Huang, Yunyi Shen, Dennis Wei, and Tamara Broderick. Dropping just a handful of preferences can change top large language model rankings. In *International Conference on Learning Representations*, volume 2026, pp. 90685–90721, 2026.

Meena Jagadeesan, Michael Jordan, Jacob Steinhardt, and Nika Haghtalab. Improved bayes risk can yield reduced social welfare under competition. *Advances in Neural Information Processing Systems*, 36:66940–66952, 2023.

Nathanael Jo, Nikhil Garg, and Manish Raghavan. The subjectivity of monoculture. *arXiv preprint arXiv:2602.24086*, 2026.

Elliot Kim, Avi Garg, Kenny Peng, and Nikhil Garg. Correlated errors in large language models. *arXiv preprint arXiv:2506.07962*, 2025.

Jon Kleinberg and Manish Raghavan. How do classifiers induce agents to invest effort strategically? *ACM Transactions on Economics and Computation (TEAC)*, 8(4):1–23, 2020.

Jon Kleinberg and Manish Raghavan. Algorithmic monoculture and social welfare. *Proceedings of the National Academy of Sciences*, 118(22):e2018340118, 2021.

Kai A Konrad and Dan Kovenock. Multi-battle contests. *Games and Economic Behavior*, 66(1): 256–274, 2009.

Anton Korinek and Jai Vipra. Concentrating intelligence: scaling and market structure in artificial intelligence. *Economic Policy*, 40(121):225–256, 2025.

Marc Lanctot, Kate Larson, Yoram Bachrach, Luke Marris, Zun Li, Avishkar Bhoopchand, Thomas Anthony, Brian Tanner, and Anna Koop. Evaluating agents using social choice theory. *arXiv preprint arXiv:2312.03121*, 2023.

Smitha Milli, John Miller, Anca D Dragan, and Moritz Hardt. The social cost of strategic classification. In *Proceedings of the conference on fairness, accountability, and transparency*, pp. 230–239, 2019.

Rui Min, Tianyu Pang, Chao Du, Qian Liu, Minhao Cheng, and Min Lin. Improving your model ranking on chatbot arena by vote rigging. *arXiv preprint arXiv:2501.17858*, 2025.

Antonio Osorio. The lottery blotto game. *Economics Letters*, 120(2):164–166, 2013.

Kenny Peng and Nikhil Garg. Monoculture in matching markets. *Advances in Neural Information Processing Systems*, 37:81959–81991, 2024.

Juan Perdomo, Tijana Zrnic, Celestine Mendler-Dünner, and Moritz Hardt. Performative prediction. In *International conference on machine learning*, pp. 7599–7609. PMLR, 2020.

Manish Raghavan. Competition and diversity in generative ai. *arXiv preprint arXiv:2412.08610*, 2024.

Arun Rajkumar, Suprovat Ghoshal, Lek-Heng Lim, and Shivani Agarwal. Ranking from stochastic pairwise preferences: Recovering condorcet winners and tournament solution sets at the top. In *International Conference on Machine Learning*, pp. 665–673. PMLR, 2015.

Pedro Rodriguez, Joe Barrow, Alexander Miserlis Hoyle, John P Lalor, Robin Jia, and Jordan Lee Boyd-Graber. Evaluation examples are not equally informative: How should that change nlp leaderboards? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 4486–4503, 2021.

Avner Shaked and John Sutton. Relaxing price competition through product differentiation. *The review of economic studies*, 49(1):3–13, 1982.

Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D'souza, Sayash Kapoor, Ahmet Üstün, Sanmi Koyejo, Yuntian Deng, Shayne Longpre, Noah A Smith, et al. The leaderboard illusion. *arXiv preprint arXiv:2504.20879*, 2025.

Stergios Skaperdas. Contest success functions. *Economic theory*, 7(2):283–290, 1996.

Gordon Tullock. Efficient rent seeking. In *Efficient rent-seeking: Chronicle of an intellectual quagmire*, pp. 3–16. Springer, 2001.

A Win Rate and ELO

We discuss the relationship between the win-rate objective in our model and Bradley–Terry/Elo-style rankings. Recall that firm i's objective depends on its pairwise probability of beating other firms across categories. In category k, the probability that firm i beats firm j is

$$p_{ij,k} = \sigma\left(F_{i,k}(\mathbf{x}_i) - F_{j,k}(\mathbf{x}_j)\right).$$

When σ is logistic, this corresponds exactly to a Bradley–Terry model within each category, where $F_{i,k}(\mathbf{x}_i)$ is the latent performance of firm i in category k. Elo uses the same logistic relationship between differences in latent scores and pairwise win probabilities, up to a rescaling of the scores.

Our model aggregates these pairwise win probabilities across categories. For category weights $\mathbf{w}$, the overall win probability of firm i against firm j is

$$p_{ij} = \sum_{k=1}^{K} w_k \sigma\left(F_{i,k}(\mathbf{x}_i) - F_{j,k}(\mathbf{x}_j)\right).$$

Thus, the objective in our model corresponds to firms maximizing a linear function of their category-level win rates. This directly captures settings in which models are evaluated according to overall win rate, or according to a weighted aggregation of category-level performance.

This is distinct from fitting a single Bradley–Terry coefficient to each model using comparisons pooled across categories. In general,

$$\sum_{k=1}^{K} w_k \sigma(z_k) \neq \sigma\left(\sum_{k=1}^{K} w_k z_k\right),$$

since a weighted average of logistic functions is not itself generally logistic. Thus, when production and performance are inherently category-specific, there need not exist a single Bradley–Terry score whose implied pairwise probabilities coincide with the weighted win probabilities in our model. A leaderboard based on a single Bradley–Terry/Elo score therefore imposes an additional one-dimensional representation of model performance. Our model instead directly captures the underlying category-level win probabilities and firms' incentives to improve their aggregate win rate.

B Situations Where an Equilibrium Doesn't Exist

We note that an equilibrium is not guaranteed to exist if σ is close to the step function. For example, suppose that

$$f_i(x_i) = x_i, \qquad c(x_i) = \alpha x_i, \qquad \sigma_\tau(z) = \frac{1}{1+e^{-\tau z}}, \qquad \tau > 4\alpha.$$

Here, $u_i(x_i; x_j) = \sigma_\tau(x_i - x_j) - \alpha x_i$, where τ controls how sharply the evaluator distinguishes between firms; as $\tau \to \infty$, σ_τ approaches a step function.

We first show that no symmetric candidate equilibrium, $x_1 = x_2 = x$ exist when $\tau > 4\alpha$. At such a profile,

$$\frac{\partial u_i}{\partial x_i} = \sigma_\tau'(0) - \alpha = \frac{\tau}{4} - \alpha > 0.$$

Therefore, both firms prefer to positively deviate, showing that such an equilibrium is not stable.

Next, we demonstrate that an asymmetric equilibrium cannot exist either. Suppose without loss of generality that $x_1 > x_2$. When a firm is behind, σ_τ is strictly convex, so the losing firm cannot have an interior optimum below its competitor. Therefore, any asymmetric equilibrium must take the form $(x, 0)$ for some $x > 0$. Since x is optimal for the winning firm against zero, it satisfies

$$u_1'(x; 0) = \sigma_\tau'(x) - \alpha = 0.$$

Moreover, since $\tau > 4\alpha$, we have $u_1'(0; 0) = \tau/4 - \alpha > 0$. Since $\sigma_\tau'(y)$ is strictly decreasing for $y > 0$, it follows that $u_1'(y; 0) > 0$ for every $y \in [0, x)$. Hence

$$u_1(x; 0) = \sigma_\tau(x) - \alpha x > u_1(0; 0) = \frac{1}{2}.$$

We now show that such a solution cannot be an equilibrium. If firm 2 deviates from 0 to $2x$, its gain is

$$u_2(2x; x) - u_2(0; x) = \sigma_\tau(x) - 2\alpha x - \sigma_\tau(-x) = 2\left(\sigma_\tau(x) - \alpha x - \frac{1}{2}\right) > 0,$$

where we use $\sigma_\tau(-x) = 1 - \sigma_\tau(x)$. Thus, the losing firm strictly prefers to leapfrog the winner, so $(x, 0)$ cannot be an equilibrium., and so no pure-strategy Nash equilibrium exists. Intuitively, when τ is sufficiently large, the benefit from marginally outperforming one's competitor exceeds the marginal cost of additional investment.

C Name Matching Details

For our usage analysis, we match model names between OpenRouter and LMArena, which use different naming conventions for provider prefixes, routing variants, release dates, and model versions. We first canonicalize names by removing provider and routing information and matching models based on their alphabetic and numeric components. We then allow matches that differ only in date stamps, and manually verify remaining models with material token volume. We deliberately avoid general fuzzy matching, since incorrect matches would attribute token volume to the wrong capability profile. Overall, we match 192 of 347 OpenRouter model slugs, covering 83.5% of pooled token volume and an average of 79.2% of monthly token volume. Models without a defensible LMArena counterpart, including pre-release codenames and models not separately evaluated by LMArena, are left unmatched. We treat unmatched token volume as falling outside any threshold of the top-rated model, making our estimates of usage near the frontier conservative.

D Open Source Models

For our scaling analysis, we combine capability scores from the Open LLM Leaderboard with parameter counts from the corresponding Hugging Face repositories. We use the leaderboard's six capability dimensions—IFEval, BBH, MATH Level 5, GPQA, MuSR, and MMLU-Pro—and standardize each dimension across the full leaderboard before restricting the sample. Because most leaderboard organizations upload fine-tunes of models produced elsewhere, we restrict our analysis to organizations with sufficient variation in model size to estimate a meaningful scaling relationship, requiring at least 7 models, 4 distinct parameter counts, and a four-fold range in parameter count. This results in 15 organizations and 293 leaderboard observations, with our specifications controlling for whether a model is instruction-tuned. Parameter count provides an observable but imperfect proxy for investment, since it does not capture differences in training tokens, compute, or post-training. Moreover, because parameter counts are generally unavailable for closed frontier models, this analysis is necessarily based on open-weight models and should not be interpreted as directly estimating the production functions of frontier firms.

E Proofs of Equilibria and Monoculture Results

E.1 Common Cost Function

Proposition 7 (Common Cost Functions)**.** *Suppose each firm i has continuous production function F_i and cost function c_i, where each $c_i : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is a strictly increasing bijection with $c_i(0) = 0$. Let c be any other such cost function. Then, for each firm i, there exists a production function $\tilde{F}_i$ and a bijection between action vectors $\mathbf{x}_i \leftrightarrow \mathbf{z}_i$ such that*

$$c_i\big(\|\mathbf{x}_i\|_1\big) = c\big(\|\mathbf{z}_i\|_1\big) \qquad \textit{and} \qquad F_i(\mathbf{x}_i) = \tilde{F}_i(\mathbf{z}_i).$$

Consequently, the game defined by $(F_i, c_i)_{i=1}^N$ is equivalent under reparameterization to the game defined by $(\tilde{F}_i, c)_{i=1}^N$: corresponding strategy profiles induce identical payoffs, and their Nash equilibria are in bijective correspondence.

Proof. Define the scalar map

$$g_i(T) := c_i^{-1}\big(c(T)\big), \qquad T \geq 0.$$

Since c_i and c are strictly increasing bijections of $\mathbb{R}_{\geq 0}$, so is g_i. In particular, g_i^{-1} exists and is itself a strictly increasing bijection of $\mathbb{R}_{\geq 0}$. By construction,

$$c_i\big(g_i(T)\big) = c(T) \qquad \text{for all } T \geq 0. \tag{1}$$

Equivalently, substituting $g_i^{-1}(T)$ for T gives

$$c\big(g_i^{-1}(T)\big) = c_i(T) \qquad \text{for all } T \geq 0. \tag{2}$$

For $\mathbf{x}_i \neq \mathbf{0}$ define the *reference-unit* vector

$$\mathbf{z}_i := \frac{g_i^{-1}\big(\|\mathbf{x}_i\|_1\big)}{\|\mathbf{x}_i\|_1}\,\mathbf{x}_i,$$

and set $\mathbf{z}_i = \mathbf{0}$ when $\mathbf{x}_i = \mathbf{0}$. Thus $\mathbf{x}_i$ is rescaled, keeping its direction fixed, so that $\|\mathbf{z}_i\|_1 = g_i^{-1}(\|\mathbf{x}_i\|_1)$. Applying g_i to both sides of this identity gives

$$g_i\big(\|\mathbf{z}_i\|_1\big) = \|\mathbf{x}_i\|_1.$$

Moreover, by (2),

$$c\big(\|\mathbf{z}_i\|_1\big) = c\big(g_i^{-1}(\|\mathbf{x}_i\|_1)\big) = c_i\big(\|\mathbf{x}_i\|_1\big),$$

so cost is exactly preserved.

For $\mathbf{z} \neq \mathbf{0}$, define

$$\tilde{F}_i(\mathbf{z}) := F_i\left(\frac{g_i\big(\|\mathbf{z}\|_1\big)}{\|\mathbf{z}\|_1}\,\mathbf{z}\right),$$

and let $\tilde{F}_i(\mathbf{0}) = F_i(\mathbf{0})$. Evaluating at $\mathbf{z} = \mathbf{z}_i$ for $\mathbf{x}_i \neq \mathbf{0}$ using $g_i\big(\|\mathbf{z}_i\|_1\big) = \|\mathbf{x}_i\|_1$ gives,

$$\tilde{F}_i(\mathbf{z}_i) = F_i\left(\frac{\|\mathbf{x}_i\|_1}{\|\mathbf{z}_i\|_1} \cdot \frac{g_i^{-1}(\|\mathbf{x}_i\|_1)}{\|\mathbf{x}_i\|_1}\,\mathbf{x}_i\right) = F_i(\mathbf{x}_i),$$

where we used $\|\mathbf{z}_i\|_1 = g_i^{-1}(\|\mathbf{x}_i\|_1)$. If $\mathbf{x}_i = \mathbf{0}$, then $\mathbf{z}_i = \mathbf{0}$ and the equality follows directly from $\tilde{F}_i(\mathbf{0}) = F_i(\mathbf{0})$. Thus $F_i(\mathbf{x}_i) = \tilde{F}_i(\mathbf{z}_i)$ as required.

Finally, the map $\mathbf{x}_i \mapsto \mathbf{z}_i$ is a bijection of $\mathbb{R}^K_{\geq 0}$, with inverse

$$\mathbf{x}_i = \frac{g_i(\|\mathbf{z}_i\|_1)}{\|\mathbf{z}_i\|_1}\mathbf{z}_i$$

for $\mathbf{z}_i \neq \mathbf{0}$, and $\mathbf{x}_i = \mathbf{0}$ when $\mathbf{z}_i = \mathbf{0}$.

Applying this construction to every firm i, corresponding strategy profiles $(\mathbf{x}_1, \ldots, \mathbf{x}_N)$ and $(\mathbf{z}_1, \ldots, \mathbf{z}_N)$ induce identical outputs and costs for every firm, and hence identical payoffs. Since the correspondence is a bijection for each firm, best responses are preserved under the reparameterization, and therefore the Nash equilibria of the two games are in bijective correspondence. □

E.2 Proof of Theorem 1

Theorem 1 (Equilibrium Existence)**.** *Suppose that each $g_{i,k}(\mathbf{x}_i) = [F_i(\mathbf{x}_i)]_k$ is twice continuously differentiable, c is convex, strictly increasing, continuous, and unbounded with $c(0) = 0$, and σ is twice continuously differentiable with $\sigma'(z) > 0$. Let*

$$\rho_\sigma := \sup_{z \in \mathbb{R}} \frac{\sigma''(z)}{\sigma'(z)}.$$

If

$$-\nabla^2 g_{i,k}(\mathbf{x}_i) \succeq \rho_\sigma \nabla g_{i,k}(\mathbf{x}_i) \nabla g_{i,k}(\mathbf{x}_i)^\top \qquad \forall\, \mathbf{x}_i,\ \forall\, i, k,$$

then the game admits a pure-strategy Nash equilibrium.

Proof. Since σ is bounded in $[0, 1]$ and $\mathbf{w} \in \Delta^{K-1}$, the total reward of any firm is at most 1. Moreover, choosing $\mathbf{x}_i = \mathbf{0}$ gives nonnegative utility. Therefore, any action satisfying

$$c(\|\mathbf{x}_i\|_1) > 1$$

cannot be a best response. Let $\bar{x} = c^{-1}(1)$. We may therefore restrict each firm's strategy set to

$$\mathcal{X}_i = \{\mathbf{x}_i \in \mathbb{R}_+^K : \|\mathbf{x}_i\|_1 \le \bar{x}\},$$

which is compact and convex. We note that u_i is continuous because it is built from the addition and multiplication of continuous functions in σ, F_i, and c.

It remains to show that $u_i(\mathbf{x}_i, \mathbf{x}_{-i})$ is concave in $\mathbf{x}_i$. For $z_{ij,k} = g_{i,k}(\mathbf{x}_i) - g_{j,k}(\mathbf{x}_j)$,

$$\nabla^2_{\mathbf{x}_i} \sigma(z_{ij,k}) = \sigma'(z_{ij,k}) \nabla^2 g_{i,k}(\mathbf{x}_i) + \sigma''(z_{ij,k}) \nabla g_{i,k}(\mathbf{x}_i) \nabla g_{i,k}(\mathbf{x}_i)^\top.$$

Since $\sigma'(z) > 0$ and

$$\frac{\sigma''(z)}{\sigma'(z)} \le \rho_\sigma,$$

the assumed curvature condition implies

$$\nabla^2_{\mathbf{x}_i} \sigma(z_{ij,k}) \preceq 0.$$

Thus every pairwise reward term is concave in $\mathbf{x}_i$. Their nonnegative weighted sum is therefore concave, and since c is convex,

$$-c(\|\mathbf{x}_i\|_1)$$

is concave as well. Hence $u_i(\cdot, \mathbf{x}_{-i})$ is concave on $\mathcal{X}_i$ for every $\mathbf{x}_{-i}$.

Finally, the utilities are continuous and the strategy sets $\mathcal{X}_i$ are nonempty, compact, and convex. Therefore, by the Debreu–Fan–Glicksberg equilibrium existence theorem, the game admits a pure-strategy Nash equilibrium. □

E.3 Proof of Lemma 1

Proof. By Theorem 1, we may restrict each firm's action space to a compact convex set $\mathcal{X}_i$, and each payoff u_i is continuous and concave in the firm's own action. Hence each best-response correspondence is nonempty, convex-valued, and upper hemicontinuous.

First suppose that $F_i = F_{i'}$. Consider the subset of strategy profiles

$$\mathcal{X}^{i,i'} = \left\{ \mathbf{x} \in \prod_{j=1}^{N} \mathcal{X}_j : \mathbf{x}_i = \mathbf{x}_{i'} \right\}.$$

At any profile $\mathbf{x} \in \mathcal{X}^{i,i'}$, firms i and i' face identical optimization problems: they have the same production function and cost function, and, because $\mathbf{x}_i = \mathbf{x}_{i'}$, each faces the same actions and outputs from all other firms. Therefore,

$$\mathrm{BR}_i(\mathbf{x}_{-i}) = \mathrm{BR}_{i'}(\mathbf{x}_{-i'}).$$

Since these best-response sets are nonempty and convex, we may choose the same best response for firms i and i'. This restricted best-response space maps $\mathcal{X}^{i,i'}$ into itself and is nonempty, convex-valued, and upper hemicontinuous. By Katukani's fixed-point theorem, there exists some fixed point, $\mathbf{x}^* \in \mathcal{X}^{i,i'}$. This fixed point satisfies $\mathbf{x}_i^* = \mathbf{x}_{i'}^*$, by the construction of $\mathcal{X}^{i,i'}$. Because $F_i = F_{i'}$, we have that $\boldsymbol{\theta}_i = \boldsymbol{\theta}_{i'}$, and so $F_{i,i'} = 1$.

Now suppose that $F_i = F$ for every i. Now again consider the fully symmetric subset

$$\mathcal{X}^{\text{sym}} = \left\{ \mathbf{x} \in \prod_{j=1}^{N} \mathcal{X}_j : \mathbf{x}_1 = \cdots = \mathbf{x}_N \right\}.$$

Here, every firm faces the same optimization problem, and so similar to before, face the same best response problem. We can choose the same best response for each firm under this situation. As above, the resulting correspondence maps $\mathcal{X}^{\text{sym}}$ into itself and satisfies the conditions of Kakutani's fixed-point theorem. Therefore, there exists a fixed point with $\mathbf{x}_i$ and F_i constant, which implies $\boldsymbol{\theta}_i$ constant, and consequently $H = 1$. □

E.4 Proof of Lemma 2

Lemma 2 (Similarity in Production Functions)**.** *Suppose the conditions of Theorem 1 hold. Additionally, suppose that the production functions are L-Lipschitz, and that the pseudogradient G is μ-strongly monotone:*

$$(G(\mathbf{x}) - G(\mathbf{x}'))^\top (\mathbf{x} - \mathbf{x}') \geq \mu \|\mathbf{x} - \mathbf{x}'\|_2^2 \qquad \forall \mathbf{x}, \mathbf{x}'.$$

If firms i and i' have similar production functions in both value and gradient:

$$\sup_{\mathbf{x}_i \in \mathcal{X}_i} \|F_i(\mathbf{x}_i) - F_{i'}(\mathbf{x}_i)\|_2 \leq \epsilon, \qquad \sup_{\mathbf{x}_i \in \mathcal{X}_i} \|\nabla F_i(\mathbf{x}_i)_k - \nabla F_{i'}(\mathbf{x}_i)_k\|_2 \leq \epsilon \qquad \forall k.$$

then the equilibrium $\mathbf{x}^*$ satisfies

$$\|\mathbf{x}_i^* - \mathbf{x}_{i'}^*\|_2 \leq O\left(\frac{\epsilon}{\mu}\right), \qquad \|F_i(\mathbf{x}_i^*) - F_{i'}(\mathbf{x}_{i'}^*)\|_2 \leq O\left(\epsilon\left(1 + \frac{L}{\mu}\right)\right).$$

Moreover, if $\|\boldsymbol{\theta}_i^*\|_2, \|\boldsymbol{\theta}_{i'}^*\|_2 \geq m > 0$, then

$$H_{i,i'} \geq 1 - O\left(\frac{\epsilon^2}{m^2}\left(1 + \frac{L}{\mu}\right)^2\right).$$

Proof. Let $G(\mathbf{x})$ be the pseudogradient for the original game with production functions F_i and $F_{i'}$ and let $\widetilde{G}(\mathbf{x})$ be the pseudogradient for the counterfactual game where F_i is replaced with $F_{i'}$ (while everything else stays constant).

We show that G and $\widetilde{G}$ are close in L2 norm. For each firm ℓ, the corresponding block of the pseudogradient is

$$G_\ell(\mathbf{x}) = -\nabla_{\mathbf{x}_\ell} u_\ell(\mathbf{x}).$$

Since

$$u_\ell(\mathbf{x}) = \frac{1}{N-1} \sum_{r \neq \ell} \sum_{k=1}^{K} w_k \sigma(F_\ell(\mathbf{x}_\ell)_k - F_r(\mathbf{x}_r)_k) - c(\|\mathbf{x}_\ell\|_1),$$

we have

$$\nabla_{\mathbf{x}_\ell} u_\ell(\mathbf{x}) = \frac{1}{N-1} \sum_{r \neq \ell} \sum_{k=1}^{K} w_k \sigma'(F_\ell(\mathbf{x}_\ell)_k - F_r(\mathbf{x}_r)_k) \nabla_{\mathbf{x}_\ell} F_\ell(\mathbf{x}_\ell)_k - \nabla_{\mathbf{x}_\ell} c(\|\mathbf{x}_\ell\|_1).$$

Since the strategy space is compact and the production functions are continuous, the arguments of σ lie in a compact interval. Since σ is twice continuously differentiable, σ'' is bounded on this interval, and hence σ' is Lipschitz. Replacing F_i with $F_{i'}$ affects only terms involving firm i. For $\ell = i$, both the argument of σ' and the production-function gradient change; for $\ell \neq i$, only the comparison

against firm i changes. Using $\|F_i(\mathbf{x}_i) - F_{i'}(\mathbf{x}_i)\|_2 \le \epsilon$, $\|\nabla F_i(\mathbf{x}_i)_k - \nabla F_{i'}(\mathbf{x}_i)_k\|_2 \le \epsilon$, and the Lipschitz continuity of σ', we can bound these differences term-by-term to obtain

$$\sup_{\mathbf{x}\in\mathcal{X}} \|G(\mathbf{x}) - \widetilde{G}(\mathbf{x})\|_2 \le C\epsilon$$

for some constant $C > 0$ independent of ϵ.

Next, let $\mathbf{x}^*$ denote the equilibrium of the original game. Because F_i and $F_{i'}$ both satisfy the conditions of Theorem 1, the counterfactual game also satisfies these conditions. By Lemma 1, there exists an equilibrium $\widetilde{\mathbf{x}}^*$ of the counterfactual game such that $\widetilde{\mathbf{x}}^*_i = \widetilde{\mathbf{x}}^*_{i'}$. We select such an equilibrium $\widetilde{\mathbf{x}}^*$. Since $\mathbf{x}^*$ and $\widetilde{\mathbf{x}}^*$ are equilibria of the original and counterfactual games, respectively, they satisfy

$$G(\mathbf{x}^*)^\top(\widetilde{\mathbf{x}}^* - \mathbf{x}^*) \ge 0, \qquad \widetilde{G}(\widetilde{\mathbf{x}}^*)^\top(\mathbf{x}^* - \widetilde{\mathbf{x}}^*) \ge 0.$$

Let $d = \mathbf{x}^* - \widetilde{\mathbf{x}}^*$. Adding the two inequalities gives $(G(\mathbf{x}^*) - \widetilde{G}(\widetilde{\mathbf{x}}^*))^\top d \le 0$. Recall that $G(\mathbf{x})$ is strongly monotone, and so

$$\left(G(\mathbf{x}) - G(\mathbf{x}')\right)^\top (\mathbf{x} - \mathbf{x}') \ge \mu\|\mathbf{x} - \mathbf{x}'\|_2^2.$$

Setting $\mathbf{x} = \mathbf{x}^*$ and $\mathbf{x}' = \widetilde{\mathbf{x}}^*$ gives

$$(G(\mathbf{x}^*) - G(\widetilde{\mathbf{x}}^*))^\top d \ge \mu\|d\|_2^2.$$

Invoking $(G(\mathbf{x}^*) - \widetilde{G}(\widetilde{\mathbf{x}}^*))^\top d \le 0$ gives that

$$\mu\|d\|_2^2 \le (G(\mathbf{x}^*) - G(\widetilde{\mathbf{x}}^*))^\top d \le (\widetilde{G}(\widetilde{\mathbf{x}}^*) - G(\widetilde{\mathbf{x}}^*))^\top d \le C\epsilon\|d\|_2,,$$

where the final inequality follows from Cauchy–Schwarz and the bound on the difference between the pseudogradients. Rearranging and solving for d gives that

$$\|\mathbf{x}^* - \widetilde{\mathbf{x}}^*\|_2 \le \frac{C\epsilon}{\mu}.$$

Since $\widetilde{\mathbf{x}}^*_i = \widetilde{\mathbf{x}}^*_{i'}$,

$$\|\mathbf{x}^*_i - \mathbf{x}^*_{i'}\|_2 \le \|\mathbf{x}^*_i - \widetilde{\mathbf{x}}^*_i\|_2 + \|\mathbf{x}^*_{i'} - \widetilde{\mathbf{x}}^*_{i'}\|_2 \le O(\|\mathbf{x}^* - \widetilde{\mathbf{x}}^*\|_2) = O\left(\frac{\epsilon}{\mu}\right).$$

Next, we use the Lipschitz-continuity of production functions to bound the difference between equilibrium outputs as

$$\|\mathbf{x}^*_i - \mathbf{x}^*_{i'}\|_2 \le \|\mathbf{x}^*_i - \widetilde{\mathbf{x}}^*_i\|_2 + \|\mathbf{x}^*_{i'} - \widetilde{\mathbf{x}}^*_{i'}\|_2 \le \sqrt{2}\|\mathbf{x}^* - \widetilde{\mathbf{x}}^*\|_2 \le \frac{\sqrt{2}C\epsilon}{\mu} = O\left(\frac{\epsilon}{\mu}\right).$$

Finally, to bound $H_{i,i'}$, let $\delta = \|\boldsymbol{\theta}^*_i - \boldsymbol{\theta}^*_{i'}\|_2 = O(\epsilon(1 + L/\mu))$. Recall that

$$H_{i,i'} = \frac{\langle \boldsymbol{\theta}^*_i, \boldsymbol{\theta}^*_{i'}\rangle}{\|\boldsymbol{\theta}^*_i\|_2\|\boldsymbol{\theta}^*_{i'}\|_2}.$$

Let $\widehat{\boldsymbol{\theta}}^*_i = \boldsymbol{\theta}^*_i/\|\boldsymbol{\theta}^*_i\|_2$ and $\widehat{\boldsymbol{\theta}}^*_{i'} = \boldsymbol{\theta}^*_{i'}/\|\boldsymbol{\theta}^*_{i'}\|_2$. Since both normalized vectors have unit norm,

$$\|\widehat{\boldsymbol{\theta}}^*_i - \widehat{\boldsymbol{\theta}}^*_{i'}\|_2^2 = 2 - 2\langle\widehat{\boldsymbol{\theta}}^*_i, \widehat{\boldsymbol{\theta}}^*_{i'}\rangle = 2(1 - H_{i,i'}).$$

If $\|\boldsymbol{\theta}^*_i\|_2, \|\boldsymbol{\theta}^*_{i'}\|_2 \ge m$, then, adding and subtracting $\boldsymbol{\theta}^*_{i'}/\|\boldsymbol{\theta}^*_i\|_2$ and applying the triangle inequality,

$$\begin{aligned}
\left\|\frac{\boldsymbol{\theta}^*_i}{\|\boldsymbol{\theta}^*_i\|_2} - \frac{\boldsymbol{\theta}^*_{i'}}{\|\boldsymbol{\theta}^*_{i'}\|_2}\right\|_2 &= \left\|\frac{\boldsymbol{\theta}^*_i - \boldsymbol{\theta}^*_{i'}}{\|\boldsymbol{\theta}^*_i\|_2} + \boldsymbol{\theta}^*_{i'}\left(\frac{1}{\|\boldsymbol{\theta}^*_i\|_2} - \frac{1}{\|\boldsymbol{\theta}^*_{i'}\|_2}\right)\right\|_2 \\
&\le \frac{\|\boldsymbol{\theta}^*_i - \boldsymbol{\theta}^*_{i'}\|_2}{\|\boldsymbol{\theta}^*_i\|_2} + \|\boldsymbol{\theta}^*_{i'}\|_2\left|\frac{1}{\|\boldsymbol{\theta}^*_i\|_2} - \frac{1}{\|\boldsymbol{\theta}^*_{i'}\|_2}\right| \\
&= \frac{\|\boldsymbol{\theta}^*_i - \boldsymbol{\theta}^*_{i'}\|_2}{\|\boldsymbol{\theta}^*_i\|_2} + \frac{|\|\boldsymbol{\theta}^*_i\|_2 - \|\boldsymbol{\theta}^*_{i'}\|_2|}{\|\boldsymbol{\theta}^*_i\|_2} \\
&\le \frac{2\|\boldsymbol{\theta}^*_i - \boldsymbol{\theta}^*_{i'}\|_2}{m} = \frac{2\delta}{m}.
\end{aligned}$$

Therefore,

$$2(1 - H_{i,i'}) \leq \frac{4\delta^2}{m^2},$$

and hence

$$H_{i,i'} \geq 1 - \frac{2\delta^2}{m^2} = 1 - O\left(\frac{\epsilon^2}{m^2}\left(1 + \frac{L}{\mu}\right)^2\right).$$

□

E.5 Proof of Lemma 3

Lemma 3 (Specialized Production Functions Induce Differentiation)**.** *Consider two firms i and i', and suppose there exist disjoint sets of dimensions $S_i, S_{i'} \subseteq [K]$ and some $\gamma \in [0, 1]$ such that*

$$\left\|F_i(\mathbf{x}_i)_{S_i^c}\right\|_2 \leq \gamma \left\|F_i(\mathbf{x}_i)_{S_i}\right\|_2 \qquad \forall \mathbf{x}_i \in \mathcal{X}_i,$$

and

$$\left\|F_{i'}(\mathbf{x}_{i'})_{S_{i'}^c}\right\|_2 \leq \gamma \left\|F_{i'}(\mathbf{x}_{i'})_{S_{i'}}\right\|_2 \qquad \forall \mathbf{x}_{i'} \in \mathcal{X}_{i'}.$$

Then any equilibrium $\mathbf{x}^*$ with $\|\boldsymbol{\theta}_i^*\|_2, \|\boldsymbol{\theta}_{i'}^*\|_2 > 0$ satisfies

$$H_{i,i'} \leq \frac{2\gamma}{1 + \gamma^2}.$$

In particular, if $\gamma = 0$, then $H_{i,i'} = 0$.

Proof. By assumption, each firm's production function primarily produces output in its corresponding set of dimensions. That is,

$$\left\|F_i(\mathbf{x}_i)_{S_i^c}\right\|_2 \leq \gamma \left\|F_i(\mathbf{x}_i)_{S_i}\right\|_2.$$

Evaluating this condition at the equilibrium $\mathbf{x}^*$, where $\boldsymbol{\theta}_i^* = F_i(\mathbf{x}_i^*)$, gives

$$\left\|\boldsymbol{\theta}_{i,S_i^c}^*\right\|_2 \leq \gamma \left\|\boldsymbol{\theta}_{i,S_i}^*\right\|_2, \qquad \left\|\boldsymbol{\theta}_{i',S_{i'}^c}^*\right\|_2 \leq \gamma \left\|\boldsymbol{\theta}_{i',S_{i'}}^*\right\|_2.$$

Next, we analyze the geometry of this situation to show that $\boldsymbol{\theta}_i^*$ and $\boldsymbol{\theta}_{i'}^*$ are nearly orthogonal. Let α_i denote the angle between $\boldsymbol{\theta}_i^*$ and its projection onto the dimensions S_i, $\boldsymbol{\theta}_{i,S_i}^*$. Because the components on S_i and S_i^c are orthogonal,

$$\tan(\alpha_i) = \frac{\|\boldsymbol{\theta}_{i,S_i^c}^*\|_2}{\|\boldsymbol{\theta}_{i,S_i}^*\|_2} \leq \gamma,$$

and therefore $\alpha_i \leq \arctan(\gamma)$. Analogously, if $\alpha_{i'}$ denotes the angle between $\boldsymbol{\theta}_{i'}^*$ and its projection onto $S_{i'}$, then $\alpha_{i'} \leq \arctan(\gamma)$.

Since $S_i \cap S_{i'} = \emptyset$, vectors supported on S_i and $S_{i'}$ are orthogonal. Therefore, the angle between $\boldsymbol{\theta}_i^*$ and $\boldsymbol{\theta}_{i'}^*$ is at least

$$\angle(\boldsymbol{\theta}_i^*, \boldsymbol{\theta}_{i'}^*) \geq \frac{\pi}{2} - \alpha_i - \alpha_{i'} \geq \frac{\pi}{2} - 2\arctan(\gamma).$$

Since $\gamma \in [0, 1]$, this lower bound is nonnegative. It follows that

$$H_{i,i'} = \cos\left(\angle(\boldsymbol{\theta}_i^*, \boldsymbol{\theta}_{i'}^*)\right) \leq \cos\left(\frac{\pi}{2} - 2\arctan(\gamma)\right) = \sin(2\arctan(\gamma)) = \frac{2\gamma}{1 + \gamma^2}.$$

□

E.6 Proof for Proposition 1

Proposition 1 (Local Compatibility is Necessary for Monoculture)**.** *For each firm i, define its set of locally supportable output directions as*

$$\mathcal{B}_i(w) = \left\{ \frac{\mathbf{d}}{\|\mathbf{d}\|_2} : \mathbf{d} \text{ is locally supportable by firm } i \right\}.$$

If there exists a pure-strategy equilibrium with complete monoculture and nonzero outputs, then

$$\bigcap_{i=1}^{N} \mathcal{B}_i(w) \neq \emptyset.$$

Proof. Suppose that $\mathbf{x}^*$ is a pure-strategy equilibrium with complete monoculture and nonzero outputs. Then there exists $\mathbf{d} \in \mathbb{S}_+^{K-1}$ and scalars $\alpha_1, \ldots, \alpha_N > 0$ such that

$$F_i(\mathbf{x}_i^*) = \alpha_i \mathbf{d} \qquad \text{for all } i \in [N].$$

For each firm i, define its marginal evaluation vector at $\mathbf{x}^*$ by

$$p_{i,k}^* = \frac{w_k}{N-1} \sum_{j \neq i} \sigma'\big(F_{i,k}(\mathbf{x}_i^*) - F_{j,k}(\mathbf{x}_j^*)\big)\,.$$

Since $F_i(\mathbf{x}_i^*) = \alpha_i \mathbf{d}$ for every i, this can be written as

$$p_{i,k}^* = \frac{w_k}{N-1} \sum_{j \neq i} \sigma'((\alpha_i - \alpha_j) d_k)\,.$$

Because $\mathbf{x}_i^*$ is optimal for firm i, the first-order conditions for the nonnegativity constraints imply

$$\nabla F_i(\mathbf{x}_i^*)^\top \mathbf{p}_i^* \leq c'(\|\mathbf{x}_i^*\|_1)\mathbf{1},$$

with equality in every coordinate ℓ for which $x_{i,\ell}^* > 0$.

Thus, $\mathbf{d}$ satisfies all conditions required for membership in $\mathcal{B}_i$ for every firm i. Hence

$$\mathbf{d} \in \bigcap_{i=1}^{N} \mathcal{B}_i,$$

and therefore

$$\bigcap_{i=1}^{N} \mathcal{B}_i \neq \emptyset.$$

□

E.7 Proof for Proposition 4

Proposition 4 (Bounded Investment Gaps)**.** *Let $\mathbf{X}_i^t \in \mathbb{R}_+^K$ denote the cumulative investment of firm i at time t. Suppose that each coordinate of F_i is increasing and concave, with $F_i(\mathbf{0}) = \mathbf{0}$.*

Suppose cumulative investment gaps remain bounded:

$$\sup_t \|\mathbf{X}_i^t - \mathbf{X}_j^t\|_\infty < \infty \qquad \textit{for every } i, j,$$

while $\min_k X_{i,k}^t \to \infty$. Suppose further that production technologies are asymptotically equivalent:

$$\frac{\|F_i(\mathbf{x}) - F_j(\mathbf{x})\|_2}{\|F_j(\mathbf{x})\|_2} \to 0 \qquad \textit{as } \min_k x_k \to \infty$$

for every pair of firms i, j. Then firms bunch in relative performance:

$$\frac{\|F_i(\mathbf{X}_i^t) - F_j(\mathbf{X}_j^t)\|_2}{\max\{\|F_i(\mathbf{X}_i^t)\|_2, \|F_j(\mathbf{X}_j^t)\|_2\}} \to 0 \qquad \textit{for every } i, j.$$

Proof. Fix two firms i and j, and let

$$C_{ij} = \sup_t \|\mathbf{X}_i^t - \mathbf{X}_j^t\|_\infty < \infty.$$

Define

$$m_t = \min_k X_{j,k}^t.$$

By assumption, $m_t \to \infty$. Hence, for all sufficiently large t,

$$\left(1 - \frac{C_{ij}}{m_t}\right) \mathbf{X}_j^t \le \mathbf{X}_i^t \le \left(1 + \frac{C_{ij}}{m_t}\right) \mathbf{X}_j^t,$$

where the inequalities hold coordinatewise.

For any coordinate k, monotonicity and concavity of $F_{j,k}$, together with $F_{j,k}(\mathbf{0}) = 0$, imply that for $\lambda \in [0,1]$ and $\gamma \ge 1$,

$$F_{j,k}(\lambda \mathbf{x}) \ge \lambda F_{j,k}(\mathbf{x}), \qquad F_{j,k}(\gamma \mathbf{x}) \le \gamma F_{j,k}(\mathbf{x}).$$

Therefore,

$$\left(1 - \frac{C_{ij}}{m_t}\right) F_j(\mathbf{X}_j^t) \le F_j(\mathbf{X}_i^t) \le \left(1 + \frac{C_{ij}}{m_t}\right) F_j(\mathbf{X}_j^t),$$

coordinatewise. Additionally, the same coordinatewise bounds imply

$$\left(1 - \frac{C_{ij}}{m_t}\right) \|F_j(\mathbf{X}_j^t)\|_2 \le \|F_j(\mathbf{X}_i^t)\|_2 \le \left(1 + \frac{C_{ij}}{m_t}\right) \|F_j(\mathbf{X}_j^t)\|_2.$$

Therefore,

$$\frac{\|F_j(\mathbf{X}_i^t)\|_2}{\|F_j(\mathbf{X}_j^t)\|_2} \to 1.$$

Now, by the triangle inequality,

$$\frac{\|F_i(\mathbf{X}_i^t) - F_j(\mathbf{X}_j^t)\|_2}{\|F_j(\mathbf{X}_j^t)\|_2} \le \frac{\|F_i(\mathbf{X}_i^t) - F_j(\mathbf{X}_i^t)\|_2}{\|F_j(\mathbf{X}_j^t)\|_2} + \frac{\|F_j(\mathbf{X}_i^t) - F_j(\mathbf{X}_j^t)\|_2}{\|F_j(\mathbf{X}_j^t)\|_2}.$$

By asymptotic equivalence,

$$\frac{\|F_i(\mathbf{X}_i^t) - F_j(\mathbf{X}_i^t)\|_2}{\|F_j(\mathbf{X}_i^t)\|_2} \to 0.$$

Since we have also shown that

$$\frac{\|F_j(\mathbf{X}_i^t)\|_2}{\|F_j(\mathbf{X}_j^t)\|_2} \to 1,$$

it follows that

$$\frac{\|F_i(\mathbf{X}_i^t) - F_j(\mathbf{X}_i^t)\|_2}{\|F_j(\mathbf{X}_j^t)\|_2} \to 0.$$

□

E.8 Proof of Proposition 3

Proposition 3 (Directional Characterization of Equilibrium Similarity)**.** *Suppose that the conditions of Theorem 1 are satisfied. Let $\sigma'(z) \in [\underline{s}, \overline{s}]$, on $z \in \mathcal{Z}$ containing all attainable output differences, with $\underline{s} > 0$.*

Define the set of marginal evaluation weights

$$\mathcal{P}(w) := \left\{ p \in \mathbb{R}_+^K : \underline{s} w_k \le p_k \le \overline{s} w_k \quad \forall k \right\}.$$

For each firm i, define its directional response set

$$\mathcal{D}_i(w) := \left\{ \frac{F_i(\mathbf{x}_i(p))}{\|F_i(\mathbf{x}_i(p))\|_2} : p \in \mathcal{P}(w),\ \mathbf{x}_i(p) \in \arg\max_{\mathbf{x}_i \in \mathcal{X}_i} \left\{ p^\top F_i(\mathbf{x}_i) - c(\|\mathbf{x}_i\|_1) \right\},\ F_i(\mathbf{x}_i(p)) \neq 0 \right\}.$$

Then, at every pure-strategy Nash equilibrium with nonzero outputs,

$$\inf_{\substack{d_i \in \mathcal{D}_i(w) \\ d_{i'} \in \mathcal{D}_{i'}(w)}} d_i^\top d_{i'} \le H_{i,i'} \le \sup_{\substack{d_i \in \mathcal{D}_i(w) \\ d_{i'} \in \mathcal{D}_{i'}(w)}} d_i^\top d_{i'}.$$

Proof. Under the conditions of Theorem 1, there exists a pure-strategy Nash equilibrium $\mathbf{x}^*$. For each firm i and dimension k, define

$$p^*_{i,k} := \frac{w_k}{N-1} \sum_{i' \neq i} \sigma'\left(\theta^*_{i,k} - \theta^*_{i',k}\right).$$

Because every equilibrium output difference lies in $\mathcal{Z}$, we have

$$\underline{s} \le \sigma'\left(\theta^*_{i,k} - \theta^*_{i',k}\right) \le \overline{s}$$

for every i, i', k. Averaging across opponents therefore gives $\underline{s} w_k \le p^*_{i,k} \le \overline{s} w_k$ and hence $p^*_i \in \mathcal{P}(w)$.

By the definition of p^*_i, the KKT conditions for firm i's equilibrium problem at $\mathbf{x}^*_i$ coincide with the KKT conditions at $\mathbf{x}^*_i$ for

$$\max_{\mathbf{x}_i \in \mathcal{X}_i} \left\{ (p^*_i)^\top F_i(\mathbf{x}_i) - c(\|\mathbf{x}_i\|_1) \right\}.$$

Under the conditions of Theorem 1, each component of F_i is concave and c is convex. Since $p^*_i \ge 0$, this auxiliary objective is concave in $\mathbf{x}_i$. Its KKT conditions are therefore sufficient for optimality, implying that

$$\mathbf{x}^*_i \in \arg\max_{\mathbf{x}_i \in \mathcal{X}_i} \left\{ (p^*_i)^\top F_i(\mathbf{x}_i) - c(\|\mathbf{x}_i\|_1) \right\}.$$

Since $p^*_i \in \mathcal{P}(w)$ and $\boldsymbol{\theta}^*_i = F_i(\mathbf{x}^*_i) \neq 0$ by assumption, the definition of $\mathcal{D}_i(w)$ implies $\frac{\boldsymbol{\theta}^*_i}{\|\boldsymbol{\theta}^*_i\|_2} \in \mathcal{D}_i(w)$.

Thus, for any two firms i and i',

$$\frac{\boldsymbol{\theta}^*_i}{\|\boldsymbol{\theta}^*_i\|_2} \in \mathcal{D}_i(w), \qquad \frac{\boldsymbol{\theta}^*_{i'}}{\|\boldsymbol{\theta}^*_{i'}\|_2} \in \mathcal{D}_{i'}(w).$$

Since

$$H_{i,i'} = \left(\frac{\boldsymbol{\theta}^*_i}{\|\boldsymbol{\theta}^*_i\|_2}\right)^\top \left(\frac{\boldsymbol{\theta}^*_{i'}}{\|\boldsymbol{\theta}^*_{i'}\|_2}\right),$$

the claimed lower and upper bounds follow immediately from the definitions of $\mathcal{D}_i(w)$ and $\mathcal{D}_{i'}(w)$. □

E.9 Proof of Theorem 2

Theorem 2 (Monoculture is More Generic than Specialization)**.** *Fix an evaluation rule (w, σ) and cost function c. Let*

$$D^* = (d^*_1, \ldots, d^*_N) \in \mathcal{Y} := (S^{K-1}_+)^N$$

denote the normalized equilibrium output directions, where $d^*_i = \frac{\boldsymbol{\theta}^*_i}{\|\boldsymbol{\theta}^*_i\|_2}$. Suppose that D^* is distributed according to a continuous density on $\mathcal{Y}$ that is bounded above and below.

Let

$$\mathcal{M} := \{(d_1, \ldots, d_N) \in \mathcal{Y} : d_1 = \cdots = d_N\}, \quad \mathcal{S} := \left\{(d_1, \ldots, d_N) \in \mathcal{Y} : d_i^\top d_{i'} = 0 \quad \forall i \neq i'\right\},$$

denote the set of complete monoculture configurations and specialization configurations respectively.

For $\epsilon > 0$, let

$$\mathcal{M}_\epsilon := \{D \in \mathcal{Y} : \operatorname{dist}(D, \mathcal{M}) \le \epsilon\}, \qquad \mathcal{S}_\epsilon := \{D \in \mathcal{Y} : \operatorname{dist}(D, \mathcal{S}) \le \epsilon\}.$$

Then, when $N \le K$, as $\epsilon \to 0$,

$$\frac{\Pr(D^* \in \mathcal{S}_\epsilon)}{\Pr(D^* \in \mathcal{M}_\epsilon)} = \Theta\left(\epsilon^{N-1}\right) \longrightarrow 0.$$

Proof. We first ask what types of restrictions monoculture and specialization impose on normalized output space, $\mathcal{Y} := (S_+^{K-1})^N$, with $\dim(\mathcal{Y}) = N(K-1)$. For complete monoculture, all firms must share a common direction, d, and so once this common direction d is chosen, the entire configuration is determined. As a result, monoculture has dimensionality $\dim(\mathcal{M}) = K-1$. Because $\operatorname{codim}(\mathcal{M}) + \dim(\mathcal{M}) = \dim(\mathcal{Y}) = N(K-1)$, we have that $\operatorname{codim}_{\mathcal{Y}}(\mathcal{M}) = (N-1)(K-1)$.

Now consider complete specialization. Since all output vectors are nonnegative, $d_i^\top d_{i'} = 0$ if and only if d_i and $d_{i'}$ have disjoint supports. Thus, every specialization configuration corresponds to a collection of nonempty, pairwise disjoint support sets $S_1, \ldots, S_N \subseteq [K]$. Suppose we fix such a collection, then each vector d_i has $|S_i| - 1$ degrees of freedom. We note that there are only finitely many such collections (of the order $(N+1)^K$), and so this does not contribute to our dimension count. For a fixed collection of supports, the corresponding specialization component therefore has dimension

$$\sum_{i=1}^{N} (|S_i| - 1) = \sum_{i=1}^{N} |S_i| - N \leq K - N.$$

There are only finitely many possible support collections, so the dimension of $\mathcal{S}$ is determined by the largest such component. When $N \leq K$, we can choose pairwise disjoint nonempty supports whose union is $[K]$, attaining equality. Hence $\dim(\mathcal{S}) = K - N$, and $\operatorname{codim}_{\mathcal{Y}}(\mathcal{S}) = K(N-1)$.

It remains to translate these codimensions into probabilities of ϵ-neighborhoods. For a subset $\mathcal{A} \subseteq \mathcal{Y}$, let

$$\mathcal{A}_\epsilon = \{D \in \mathcal{Y} : \operatorname{dist}(D, \mathcal{A}) \leq \epsilon\}.$$

ϵ-neighborhoods. For a smooth component of codimension r, there are locally r directions constrained by membership in the component. To lie within distance ϵ of it, these directions may each vary only over a region of size $O(\epsilon)$. Thus, for sufficiently small ϵ, the volume of its ϵ-neighborhood is bounded above and below by positive constants times ϵ^r. This remains true at boundaries and corners of $\mathcal{Y}$: some directions may only vary on one side, which changes the constant factor but not the order in ϵ. Since $\mathcal{M}$ and $\mathcal{S}$ consist of finitely many such components, their maximal-dimensional components determine the leading order of their ϵ-neighborhood volumes. Applying this to the maximal-dimensional strata of $\mathcal{M}$ and $\mathcal{S}$ gives

$$\operatorname{Vol}(\mathcal{M}_\epsilon) = \Theta\left(\epsilon^{(N-1)(K-1)}\right), \quad \operatorname{Vol}(\mathcal{S}_\epsilon) = \Theta\left(\epsilon^{K(N-1)}\right).$$

Let f denote the density of D^*. By assumption, there exist constants $0 < \underline{f} \leq \overline{f} < \infty$ such that, in sufficiently small neighborhoods of $\mathcal{M}$ and $\mathcal{S}$, $\underline{f} \leq f(D) \leq \overline{f}$. Hence, for $\mathcal{A} \in \{\mathcal{M}, \mathcal{S}\}$,

$$\underline{f} \operatorname{Vol}(\mathcal{A}_\epsilon) \leq \Pr(D^* \in \mathcal{A}_\epsilon) \leq \overline{f} \operatorname{Vol}(\mathcal{A}_\epsilon).$$

Therefore, the probabilities have the same asymptotic order as the corresponding volumes. It follows that

$$\frac{\Pr(D^* \in \mathcal{S}_\epsilon)}{\Pr(D^* \in \mathcal{M}_\epsilon)} = \Theta\left(\epsilon^{K(N-1)-(N-1)(K-1)}\right) = \Theta\left(\epsilon^{N-1}\right) \to 0.$$

□